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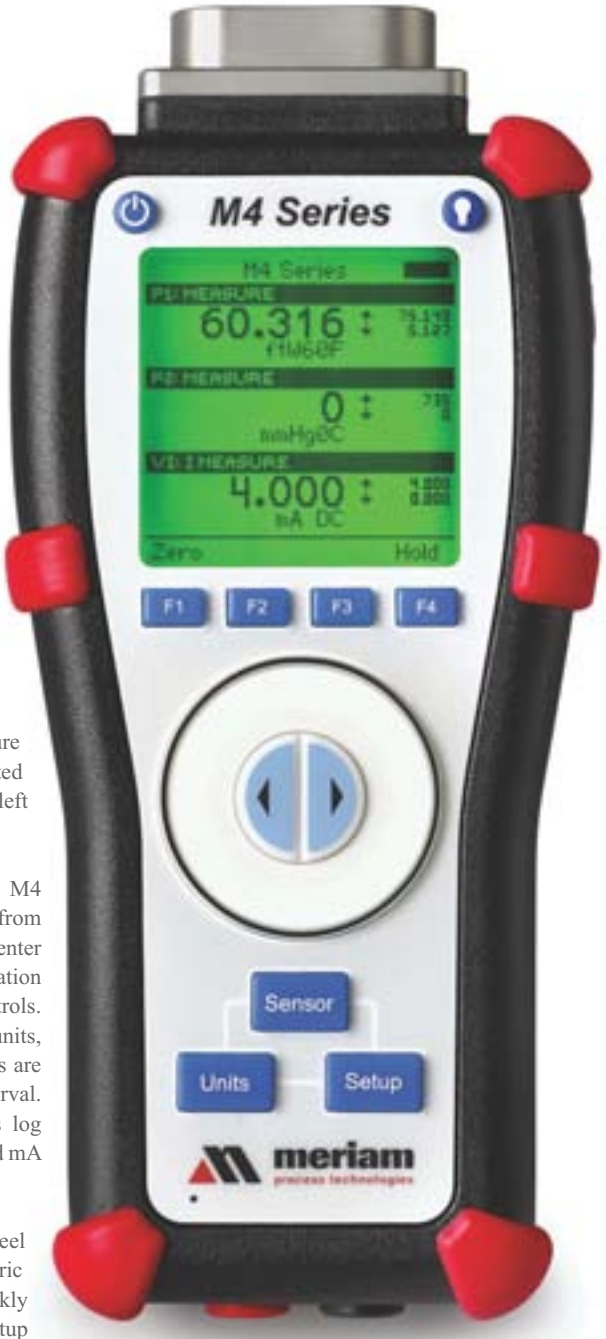
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InTech



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By Dave Adler

In pharmaceutical automation projects and beyond, define and fix your requirements. Have a robust plan, obtain management support, and maintain the discipline to execute the plan.

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Sustainability challenge

By Bill Lydon, *InTech*, Chief Editor



Sustainability is a concept I suggest should be in our thinking. In ecology, the word sustainability describes how biological systems remain diverse and productive over time. For humans, it is the potential for long-term maintenance of wellbeing, which in turn depends on the wellbeing of the natural world and the responsible use of natural resources. Some business people think of sustainability as a threat from "tree huggers" and government that will drive down profits and stifle growth. A more reasoned and productive view is to embrace the concept of sustainability to improve operations, lower costs, and improve the environment. Automation systems are an important part of achieving industrial production sustainability.

A great example of a company that embraces sustainability is PepsiCo with their "Performance with Purpose" focus. Chairman and Chief Executive Officer Indra Nooyi is clear about the goals: "Together we are all building on the platform of human, environmental, and talent sustainability while continuing to deliver great results." PepsiCo's sustainability vision is based on the high level goal, "Leave No Trace." The strategy is to conserve and preserve the earth's natural assets, particularly water, energy, and land use. PepsiCo has three strategic objectives:

- Perpetually reduce consumption of non-renewable natural assets.
- Step function change in consumer loyalty and customer intimacy.
- Embed sustainability within the cultural DNA of the company.

I had the opportunity to see a presentation by David Haft, group vice president, Sustainability & Productivity for Frito-Lay, a PepsiCo company. Haft is an engineer, and he addressed engineers at the Invensys OpsManage09 Conference describing real-world examples of how sustainability is in alignment with business results at Frito-Lay. The results are impressive: After setting goals in 1999, by 2009, they have reduced water consumption 43%, natural gas consumption 35%, and electricity 25%. Commercially in 2009, these savings were $70 million dollars or about 2.5 margin points for Frito-Lay. In addition, the company has gained recognition and awards that have marketing value.

Frito-Lay's Killingly, Conn., plant has run a combined heat and power system since 19 March 2009 to get off the power grid. Power is generated with a 6.4 megawatt gas powered turbine and the 1,000 degree Fahrenheit waste heat is used to make all the steam required for the plant. The system automatically reduces greenhouse gases by 5% by saving transmission losses, and nitrogen oxide emissions have been reduced by 60%. The system was funded in part with a more than $1 million grant from the state of Connecticut through the Energy Independence Act.

The big projects get the headline, but there are low cost projects that are low risk and have impact. Haft described a project using infrared scanners to look for heat losses from valves, steam leaks, bad steam traps, missing/bad insulation, and other energy wasters. Haft said this project was "relatively low tech but very high payback. Every point of efficiency at Frito-Lay is worth $1 million."

Success stories like this should be an inspiration and call to action for thinking creatively about what I can do to improve the sustainability of processes. Automation can be a big part of achieving sustainability to increase efficiencies or implement new functions. I suggest spending some time, may be once a week, thinking about how to improve sustainability and writing your ideas down so they can incubate and then form action plans. Collaborating with others in your operation is also productive with some opportunities requiring the coordination of multiple disciplines. It is important to clearly state the goal and potential savings to justify doing these projects.

Please share any thoughts and successes at **blydon@isa.org.**

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Determining value

Jim Pinto's pricing paradigm sounds like a win-win situation for both seller and buyer (Jan/Feb *InTech*). My question is: How can "value" be determined?

We make widgets that become part of a process control system. The sale of our products is through intermediaries. We seldom see where the products are actually used. Besides situations where, for example, IBM installs a system for a state's DMV and can charge for the number of customers processed, how would you determine value in our situation?

Maris Graube, Relcom

Response:

Performance-based pricing is typically for large systems (like DCS and SCADA systems), which need major budgeting and planning. The buyers already have a break-even analysis based on review bids. This is where performance-based pricing makes it easy—some of the price is based on future performance, which of course must be jointly evaluated based on the buyers objectives.

"Widgets" are harder to price based on performance and typically priced competitively.

The question becomes what is a "widget?" And what is a "system." The key is to talk to the customer to find out why they're buying, and what they expect. The attitude of "performance based" is itself a welcomed idea for many customers.

The key: Find what works for your company. I wish you success.

Jim Pinto



Defending Detroit

I object to the notion that the Detroit automakers "...forced cars they produced down the public's throat." I know that's not exactly what you said (September "Talk to me"), but that's the implication. The public is going to buy what it wants to buy, neither the government nor the automakers can or should do anything about that. Detroit has been slammed for making big SUVs and trucks rather than the small cars "people want." It is interesting that Ford sells three times as many F series trucks as it does the economical Focus. It is interesting that the small carmakers—Honda, Toyota, Nissan—have entered the big truck, big SUV, big luxury car markets. If anyone wants to know what the public wants, just look at what's on the road, and don't blame Detroit for making what people want to buy.

John Marshall

Our pollution footprint

Regarding the *InTech* October 2009 "The Final Say," I agree the sun is the main source of the atmospheric heating of our planet; however, I've been in the controls industry for over 40 years and have worked in most industries, petro-chemical, steel, etc.

Even as a young guy, I realized that dumping human generated waste into the environment was bad (industrial or otherwise). I worked at a plant in the U.K., which at the time dumped untreated hydrocarbon waste into the local river.

That the current drive to limit CO_2 emissions is driven by global warming fears, I for one believe it is a good thing. We should be at all times looking at what we can do to mitigate our pollution footprint on our home planet.

With regard to Mars, if it is proven that no life exists there, then by all means, use it as a laboratory.

Derek Appleton, Industrial System Arts Inc.

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Radar to map ice formations remotely



The National Aeronautics and Space Administration (NASA) has awarded $2.4 million to the Georgia Institute of Technology to develop a new type of radar system that will be used to study the Earth's ice and snow formations from the air. The system could provide new information about the effects of global climate change.

The research will create a small, lightweight, low-cost phased-array radar that uses silicon-germanium (SiGe) chips in tandem with radio-frequency micro-electromechanical systems. The system being developed could be mounted on aircraft or satellites to enable high-quality mapping of ice and snow formations.

Traditionally, research on frozen areas has required bulky radar equipment that must be operated on the surface, said John Papapolymerou, a professor in Georgia Tech's School of Electrical and Computer Engineering who is principal investigator on the project. The lightweight radar approach could allow unmanned aerial vehicles to gather information by flying over a large area such as Greenland, using the radar system to map ice sheets in three dimensions.

"This aerial approach would greatly facilitate environmental remote sensing of ice, allowing us to map larger areas of interest to better understand location, quantity and composition," said Papapolymerou, who is teamed with another Georgia Tech professor, John Cressler, and Ted Heath, a Georgia Tech Research Institute senior research scientist. "This mapping ability is very important because we need to know about ice accumulation, consistency and stability."

Phased-array radar technology uses fixed, interconnected antenna elements to send and receive multiple radar signals almost simultaneously. This approach employs a technique called phase-shifting to electronically steer the radar-signal beam.

The basic sub-array unit under development consists of a flat grid with eight antenna elements on a side—64 in all. These sub-arrays, measuring about 8.5 by 7 inches, can be combined to create a larger radar array capable of high-quality 3-D mapping.

Hydrophobic interface mimics hairs on spider

Engineering researchers have crafted a flat surface that refuses to get wet. Water droplets skitter across it like ball bearings tossed on ice.

University of Florida engineers have achieved what they label in a new paper as a "nearly perfect hydrophobic interface" by reproducing, on small bits of flat plastic, the shape and patterns of the minute hairs that grow on the bodies of spiders, according to *ScienceDaily*.

"They have short hairs and longer hairs, and they vary a lot. And that is what we mimic," said Wolfgang Sigmund, a professor of materials science and engineering.

Spiders use their water-repelling hairs to stay dry or avoid drowning, with water spiders capturing air bubbles and toting them underwater to breathe. When water scampers off the surface, it picks up and carries dirt with it, in effect making the surface self-cleaning. As such, it is ideal for some food packaging, or windows, or solar cells that must stay clean to gather sunlight, Sigmund said. Boat designers might coat hulls with it, making boats faster and more efficient.

Sigmund said he began working on the project about five years ago after picking up on the work of a colleague. Sigmund was experimenting with microscopic fibers when he turned to spiders, noted by biologists for at least a century for their water-repelling hairs.



As a scientist and engineer, he said, his natural tendency was to make all his fibers the same size and distance apart. But he learned that spider hairs are both long and short and variously curved and straight, forming a surface that is anything but uniform. He decided to try to mimic this random, chaotic surface using plastic hairs varying in size but averaging about 600 microns, or millionths of a meter.

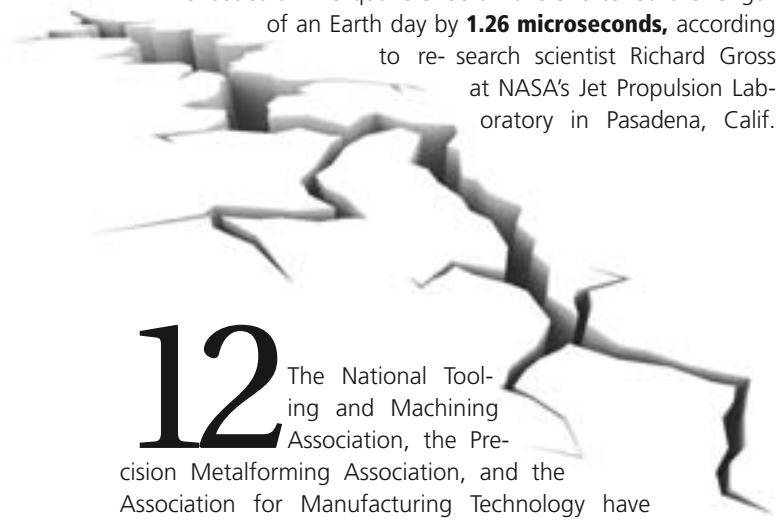
Water-repelling surfaces or treatments are already common, spanning shoe wax to caulk to car windshield treatments. However, Sigmund said the UF surface may be the most or among the most water phobic. Close-up photographs of water droplets on dime-sized plastic squares show the droplets maintain their spherical shape, whether standing still or moving. Droplets bulge down on most other surfaces, dragging a kind of tail as they move. Sigmund said his surface is the first to shuttle droplets with no tail.

Also, unlike many water-repelling surfaces, the UF one relies entirely on the microscopic shape and patterns of the material—rather than its composition. In other words, physics, not chemistry, is what makes it water repellent.

Sigmund said making the water or oil-repelling surfaces involves applying a hole-filled membrane to a polymer, heating the two, and then peeling off the membrane. Made gooey by the heat, the polymer comes out of the holes in the desired thin, randomly sized fibers. While inexpensive, it is hard to produce successful surfaces with great reliability, and different techniques need to be developed to make the surfaces in commercially available quantities and size, Sigmund said. Also, he said, more research is needed to make the surfaces hardy and resistant to damage.

Automation by the Numbers

1.26 The massive 8.8 earthquake that struck Chile in February may have changed the entire Earth's rotation and shortened the length of days on our planet, a NASA scientist said. The quake should have shortened the length of an Earth day by **1.26 microseconds,** according to re- search scientist Richard Gross at NASA's Jet Propulsion Laboratory in Pasadena, Calif. One microsecond is one-millionth of a second long. "This change should be permanent," Gross said. There is a chance the Earth's rotation could relax over time, but it is too early to tell, he said. Over the course of a year, the length of a day normally changes gradually by about one millisecond, which is 1,000 microseconds. Furthermore, geologists said the city of Concepcion was moved an estimated 10 feet west during the massive earthquake, indicated by GPS measurements taken before and after the quake by teams of researchers from universities across the Americas.



12 The National Tooling and Machining Association, the Precision Metalforming Association, and the Association for Manufacturing Technology have launched a "re-shoring" initiative aimed at documenting to large manufacturers nationwide the benefits of sourcing in the U.S., including a "Re-shoring Fair" set to take place **12 May** in Irvine, Calif. The associations said re-shoring means bringing lost manufacturing jobs back to the U.S. by uniting large manufacturers with competitive domestic suppliers. "Going local can reduce a company's total costs and offer a host of other benefits, while bringing U.S. manufacturing jobs back home," they said. The move to re-shore production has grown increasingly popular in the U.S. in the face of higher transportation and fuel costs, higher wage rates, and reject rates in developing countries, the organizations assert. For more information, visit http://tiny.cc/YU9e2.

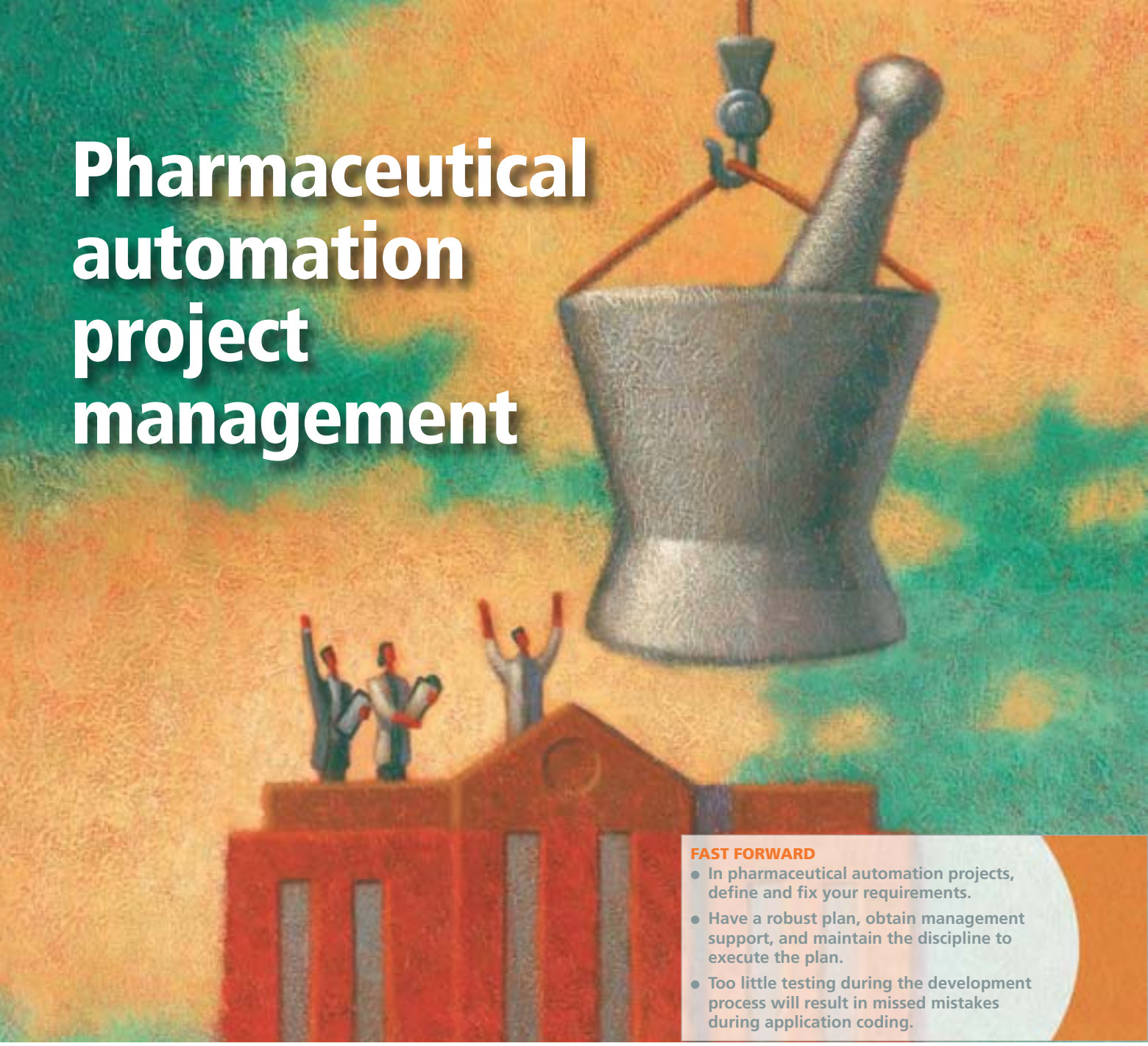
50 **Fifty years** after the first laser was demonstrated, engineers are celebrating the golden anniversary. Although there has been a historical debate over who is most properly credited as the inventor of the laser, the clearest milestone came on 16 May 1960, when Hughes Research Laboratories' Theodore Maiman demonstrated a solid-state device that used a flashlamp coiled around a ruby crystal to produce coherent pulses of red light. "Even 50 years after the invention of the laser, new applications are being patented at a phenomenal rate," said Thomas Baer, executive director of the Stanford Photonics Research Center. Patent data searches show the term "laser" ranks as the third most popular keyword, right behind "engine" and "computer."



12,500 David de Rothschild, one of the youngest members of the famous banking dynasty, wants the public to start viewing waste as a resource, particularly plastic. He and collaborators designed a boat made almost entirely of plastic bottles and recycled plastic, and in March, de Rothschild and the crew began the 11,000-mile (17,700-kilometer) voyage from San Francisco, Calif., to Sydney, Australia. The crew hopes to accomplish the voyage in 100 days on a 60-foot catamaran-style boat named The Plastiki. Builders of the boat said it weighs in at 12 tons, with only 10% of the vessel made from new materials. Constructed mainly from **12,500** reclaimed plastic water bottles designed to keep Plastiki afloat,the main frame is made from self-reinforcing polyethylene terephthalate, a recyclable plastic material, and the sail has been handmade using recycled PET cloth.



Pharmaceutical automation project management



FAST FORWARD

- In pharmaceutical automation projects, define and fix your requirements.
- Have a robust plan, obtain management support, and maintain the discipline to execute the plan.
- Too little testing during the development process will result in missed mistakes during application coding.

By Dave Adler

Did you ever wonder why it is so difficult to have a successful pharmaceutical automation project? My definition of success is measured by achieving the schedule milestones, meeting the cost estimate, satisfying the system automation requirements, having the automation system work from day one, and satisfying the facility's business leaders. To satisfy all of these measures is almost impossible. Pharmaceutical automation is tough, but when successful, it is very rewarding.

Numerous studies of software projects have found success rates of less than 20%, where success was defined as achieving schedule, meeting cost estimates, and satisfying requirements. Automation professionals who find project success challenging have lots of company. There are many ways to do automation projects poorly, but just a few ways to do them correctly.

I recently conducted research on pharmaceutical automation technology, costs, and benefits for 24 facilities of 16 member companies of the Pharmaceutical Automation Roundtable (PAR). This study analyzed the relative automation cost per input and output device (I/O). The costs per I/O varied greater than a factor of three from the least expensive to the most expensive automation system. The wide variance in automation cost per I/O in the study indicates the opportunity exists to optimize the business processes to manage automation projects.

Pharmaceutical companies have developed a

Automation is a risky endeavor—improve your odds with planning, requirements, testing, and documentation

rigorous methodology for automation systems. The industry uses a life-cycle model known as computer system validation to ensure the automation system does what it is supposed to do and can be expected to continue doing so in the future. Before I lose my non-pharmaceutical industry readers, this is just a fancy way of saying automation professionals need to:

1. Do upfront planning.
2. Define requirements for the automation system.
3. Test the automation system.
4. Document the technical content.

I hope every automation professional does each of these activities on every project, but of course not to the level of the pharmaceutical industry.

The pharmaceutical industry is regulated by the U.S. Food and Drug Administration (FDA) and the international Ministry's of Health. These regulations apply to the manufacturing of drugs and medical devices including the use of computers to manufacture these products. In 1983, the FDA published its first guide to computer system validation. Since then, the industry's automation professionals have developed the business processes to support a cradle to grave life-cycle approach to automation. The industry has had a lot of opportunity over the years to use business processes to support the delivery of automation. An industry trade group, International Society of Pharmaceutical Engineers has produced a reference guide to Good Automation Manufacturing Practice that highlights one approach widely adopted.

I have been involved with more than 20 major pharmaceutical automation projects in my career, so I have had the opportunity to be on many critical and even a few troubled automation projects. I have learned many painful lessons and now have many stories to tell. These lessons are applicable to an automation professional in any industry. Your chances of having a successful automation project can be greatly increased by using appropriate planning, requirements, coding, testing, and documentation practices.

Planning can be guide to success

Step one in improving your odds of a successful automation project is developing a plan and getting all the key stakeholders to buy into your approach. I hope by now I have convinced you how difficult it is to have a successful automation project. A structured approach, starting with a plan, can increase your odds of success.

An automation plan at a high level defines: the project drivers, the scope of work, the automation system's desired functionality, the operational strategy, the safety expectations, the maintenance strategy, the schedule, and the cost estimate. In the pharmaceutical industry, there is a regulatory requirement to have a validation master plan. An automation validation master plan defines at a high level the expectations for quality, requirements, testing, documentation, review, and approval. It would also cover expectations for security, change control, contingency planning, and periodic reviews.

There are a number of guides available to help organize a plan. Business processes are available for project managers such as those documented by the Project Management Institute that have been used by automation professionals for our discipline. Guides are available for scoping and estimating automation projects from the author.

During the initial planning of a proposed automation upgrade project, controlling cost was identified as the number one issue with getting approval. The initial proposal by the facility planning group was rejected by the management team based on the cost of other recent upgrade projects. The automation team was asked to significantly reduce the estimated cost for the proposed project. The planning effort required the automation team to think outside the box. None of the existing automation business processes were immune from review. The planning process took several months. A plan was developed that reduced the proposed automation estimate by 25%. This cost reduction was due to changes in the business process and not the overall scope of the work (e.g., fewer control loops). Highlights of the planning process were to:

1. Choose experienced handpicked automation professionals.
2. Dedicate automation staff with no other responsibilities.
3. Co-locate all automation staff in one room.
4. Ensure tech service, process engineers, and operations personnel availability when needed.
5. Define roles and responsibilities.
6. Develop a prototype for the entire software development process.
7. Replicate from previous projects.
8. Have well-defined scope and fixed requirements.
9. Create a "just say no" list of cost enhancers.

The facilities business leaders agreed and bought into this plan because they wanted the lower cost. The automation upgrade project was approved. I will skip forward several months as this project was recently completed. The automation project actually pleased the facility's business leaders. The software and hardware worked flawlessly during start-up, the project was completed on schedule, and the final automation project cost was 20% less than the revised project plan estimate. If you have a robust plan, obtain management support, and maintain the discipline to execute the plan, you can achieve your targets.

If you have a robust plan, obtain management support, and maintain the discipline to execute the plan, you can achieve your targets.

Define, fix your requirements

You must accurately define your automation project requirements with the help of the users of the automation system. The requirements should be measurable and testable. They need to identify the business, equipment, and process needs. Define the requirements before you start the design, and get the key stakeholders to agree to them. Communicate broadly these requirements. You need to keep the requirements fixed during the course of the project design, implementation, and through start-up. If you can minimize scope creep, you have removed a major hurdle to automation project success. Scope creep can result in changes and additions to requirements that can greatly lengthen and increase the cost of the project.

One of my early failures was a major pharmaceutical plant upgrade in the early 1990s that was a fast track project. It looked like we would not make enough material to launch this projected blockbuster product. The project manager and manufacturing executives wanted automation to get started and get off the critical path of the overall project. The automation team felt a lot of pressure to get started. We started ordering instruments without piping and instrument drawings. We started writing automation application code without agreed to valid requirements. In fact, we were still arguing about requirements during start-up. Not surprisingly, this project went poorly, and start-up did not go well. Cost estimates were missed, and schedule milestones were not met. Product was not made on time. This story does not have a happy ending. Even though it was not a good experience for me, I learned some valuable lessons.

Later in my career, in the mid 2000s, I was again assigned to a fast-track pharmaceutical upgrade project. The business drivers mandated a compressed schedule. The project manager and manufacturing executives wanted automation to get started and get off the critical path of the overall project. The automation team felt a lot of pressure to get started. However, we refused to write code and order instruments until we had requirements and piping and instrument drawings. Of course, I was getting worried looks and phone calls from everyone in management. We took two months to define requirements and locked in the piping and instrument drawings before we started writing code and ordering instruments. I will make a long and grueling story short and jump to the end of the story: Automation was done on schedule and on budget, and it worked well. We had a successful start-up, and within six months, automation facilitated some dramatic improvements in the operations of the facility. It does pay to plan your project and get the requirements right before you start your work.

Testing reduces start-up issues

If a developer does too little testing during the development process, there will be mistakes in the process control application code. The software will then have to be debugged later in the development process or during start-up of the manufacturing equipment in the facility. It is significantly more costly to debug software during start-up than during the development process. The sooner a developer catches a mistake, the cheaper it is to fix. Appropriate testing can reduce overall project cost and minimize rework during start-up. Of course, inappropriate testing will increase cost and lengthen the development time.

Testing determines if the automation system meets the previously defined requirements. The success of testing depends, in part, on good requirements. If a process automation professional has solid and fixed requirements, it is much easier testing.

The development process consists of testing the software at each stage of the process. The testing starts with the individual software module, and then the units should be individually tested. This module and unit testing uses "test scripts" that test small portions of the whole system software. It is important to define test scripts so the observed results are clear and concise. This testing is frequently conducted in an off-line or development system.

Once all the individual units have been tested, the overall system is tested as a whole. A portion of this may be achieved as a Factory Acceptance Test, or FAT. Final testing in the pharmaceutical industry is often defined as on-site acceptance testing and includes installation qualification, operation qualification, and performance qualification.

During a recent retrofit project, it was critical to minimize the time the facility was down. The facility was producing a life-saving medicine. The needed production output would not allow for an extended downtime. This retrofit required software and hardware changes to replace an existing obsolete automation system, including its I/O with a new automation system. This led to significant off-line testing of the automation software. In addition, a plan was developed to optimize hardware changeover. Off-line testing of the hardware was desired. An I/O room was built of plywood and plastic sheets to simulate the hardware changeover process. This room was a full-scale mock-up of the actual I/O room including the elevated floors. It allowed training of the electricians and trial runs



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of wiring efforts. The electricians practiced disassembly of the old I/O system and reassembly of the new I/O system. It allowed optimization of the number and activities of the electricians, pre-labeling terminations, fabrication of panels, and even allowed some pre-cut wire to be made up to exact lengths. This reduced the wiring effort during construction by over 75%. The actual hardware change-out took days rather than the normal weeks to change out a large I/O system. The most surprising outcome was the project actually was less expensive than planned because start-up went so well with almost no hardware and software issues. Even with the extra testing expense, this project was less expensive overall than other retrofit projects being done at the same time.

Long-term viability

Documentation is critical to the success of an automation project. If the development team does too little, the long-term support team will have a difficult time and exert additional effort to maintain the automation system. If the development team does too much, the project could be delayed and experience cost overruns.

When I first started in process automation in the early 1980s, I learned a painful lesson. On one of my first projects, I did the minimum level of documentation. It resulted in frequent phone calls from me to explain to my replacement what the design or software was doing. Since the new engineer could not figure out what I did, I clearly had not documented the automation system well. In future projects, I spent a lot more time and effort to document as I designed and coded the automation applications.

By the late 1980s, the pharmaceutical industry was implementing computer system validation. One of the unique features of the pharmaceutical industry is the actual medicine you take sometimes cannot be completely tested. A sampling of medicine from each batch is tested, but the testing itself often destroys the product. Obviously the pill that the patient takes cannot undergo a destructive test. This has led to a business process to ensure quality called *validation*. Quality has to be built into the process of manufacturing the product and the software to control the equipment. Quality cannot be tested into a product, but it must be built into the process for making the product. Industries' response to build quality into the manufacturing process resulted in progressively more rigorous and larger computer system validation documentation packages on pharmaceutical projects. By the early 2000s, automation projects were being implemented with only 10% of the effort on design and coding, but 90% of the effort on testing and producing documentation.

By the mid 2000s, the pharmaceutical industry was taking a look at using a more appropriate level of documentation. Numerous FDA investigators made comments that it was not the thickness of the documentation that was important, but rather the business value of the documentation. The FDA encouraged industry to take a critical look at its business processes to ensure cost-effective drugs for its patients. Concepts such as using a "risk-based" approach to the



overall validation process and the use of in-line quality measurements through Process Analytic Technology were initiated. This openness by the regulators to change resulted in the automation discipline taking a critical look at all its business processes including documentation.

Before the start of a major project in 2006, the level of documentation was identified as important to a cost-effective and on-schedule project. The area's management team set up an improvement team to study the computer system validation process. It defined the problem, measured the process, analyzed the data, improved the process, and set up a control system to manage the new process. The process completely prototyped the workflows and insured they functioned as intended. The process revised policies and procedures appropriately. Specific examples of improvements were using checklists instead of detailed narratives to aid in document assembly and preparation, smaller test scripts, minimizing the process to fix script errors, reducing the number of reviewers and approvers, shorter targeted QC reviews, moving to an electronic document management system from a paper system, and reducing the level of documents to support unit level testing with more focus on system level testing documentation. These efforts reduced the total level of effort on documentation by 50%, but more importantly, it produced more useful content for the long-term system support and the future optimization of the manufacturing process.

The details do matter

Contrary to a prevalent management myth that claims you do not need to worry about the details, in pharmaceutical automation projects, it is the details that really matter. If you do not worry about the details, your project will not be successful. When you work on a project you need to sweat the details including: doing upfront planning, having well defined and fixed requirements, conducting testing, and documenting the appropriate technical content.

ABOUT THE AUTHOR

Dave Adler (davidadler@comcast.net) is an automation consultant with Brillig Systems. His interests are managing automation projects and programs, developing automation strategies, developing & training automation professionals, and educating business leaders on the life-cycle cost and benefits of automation. He spent 33 years at Eli Lilly and Company in a wide variety of automation assignments. He is also currently leading ISA's workforce development efforts.

View the online version at www.isa.org/intech/20100401.

RESOURCES

***Automation Applications in Bio-Pharmaceuticals*, ISA, 2008**
www.isa.org/link/AABP_bk

A Guide to the Project Management Body of Knowledge
www.isa.org/link/PMBOK_bk

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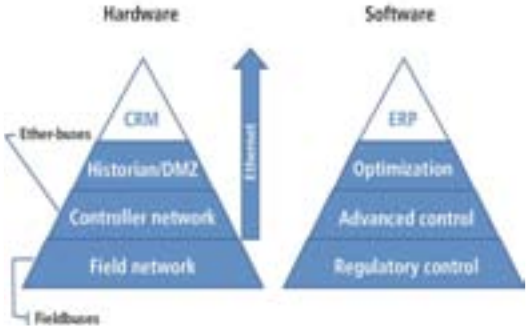
A control system is only as good as its infrastructure

By Ian Verhappen



Digital communications are becoming all pervasive and certainly in non-industrial settings are now almost being taken for granted with wireless Ethernet hot spots everywhere, PDAs/cell phones in practically every pocket or purse, and digital communications/microprocessors incorporated in a plethora of other everyday products. Despite this, the adoption of similar technologies in the automation sphere has been slower than expected based on the otherwise widespread adoption of digital communications. Part of the reason may be the lack of a "killer application" in the industrial setting, or perhaps it is simply the culture of "the existing system provides me the information I need to run my plant, so why change to something new and unproven such as fieldbus?" The reason to make the change is the "opportunity lost." A digital system provides the foundation on which significant incremental opportunities to improve facility operations can be made. Of course "opportunity lost" is difficult to quantify—sort of like buying insurance; more of a risk management item than real dollars that I lost because something happened. So why is the fact that you are not taking advantage of the digital communications in your plant a lost revenue opportunity? The reason is no different than what these systems enable, so let's see why.

Digital systems include the hardware and associated software, and the benefits of one are not possible without the other. In fact, there are some similarities between the network communications used at the various levels of the enterprise with the complexity of the network and the associated types of applications being implemented at each of the ISA95 levels to maximize the efficiency of plant operations.



Control's pyramids

As we rise "higher" in the hierarchy of control, the amount of data, number of variables, data processing requirements, and complexity of the associated software to optimize the return on capital increases almost exponentially. However, what remains true in the hardware and software realm is like all control algorithms, or any assembly for that matter, the result is only as good as its foundation. For control algorithms, this is the base regulatory controllers and associated loop tuning, and in the case of the control system hardware that foundation is the field level/fieldbus sensor and signal network.

Since the lowest layers of the hardware pyramid are most critical, the balance of this article will focus on these lower two layers. Therefore, what are considerations that must be made to provide relay signals from these levels of the control system?

Field network layer

The most widely installed digital communications protocol in process automation is HART. There are millions of HART devices installed in the world, yet more than 80% of the time, the digital capabilities of the device are not being used. Why?

Despite being able to support multi-drop communications, practically all HART installations use a point-to-point connection. The HART protocol requires that devices must be polled for any digital information, therefore it is inherently slower than other "true" fieldbuses. However, because the process data is provided as an analog signal, does the polling frequency really matter for information other than control data? This is a question that each site/operator must answer and then design their system accordingly.

FAST FORWARD

- Approximately 80% of installed smart field devices are being underutilized.
- Having the right infrastructure enables better control and higher return on investment.
- Continuing developments in hardware and software will enable better access and communication with smart field devices.

The majority of today's control systems support HART communications on their analog I/O cards, however many "legacy systems" require installation of signal "strippers" so only the pure analog signal is received and processed by the I/O card. These stripper systems use an associated parallel data gathering system, typically a combination of RS-485 and Ethernet, to a dedicated server where the information is processed as part of an asset management system.

All HART devices can also be communicated with on a one-to-one basis using handheld communicators/laptops and foregoing the asset management system—though doing so circumvents the ability to effectively mine the device diagnostic information for trends occurring across a similar range of products, or in a specific application to help you find the root cause of failures, or as a minimum, frequent "bad actors" to minimize your maintenance budget impact.

One possible reason organizations are not using the HART data they have installed in an organized manner is doing so requires a change in culture. Some technicians are afraid that by connecting their handheld units for synchronization with a server, the data collected will be used to see how much work is being completed by each of them with associated feeling of Big Brother watching. The result is all the data is on a local laptop but not being analyzed to provide the benefits of a complete asset management system, including integration with the plant work order/planning system.

Though not as common—at least not yet—full digital fieldbus systems provide the benefits of supporting multi-drop capabilities and hence multiple devices on a single network. In the wet process industries, the two most commonly used fieldbus networks are Foundation Fieldbus (www.fieldbus.org) and Profibus PA (www.profibus.com), both using the same physical layer of individually shielded twisted pair cables wired in parallel and a Manchester encoded 31.25 kbps signal.

Both of these standards included as part of their design basis reuse of existing infrastructure and full backwards compatibility with previous versions of the protocol from revision one to infinity (whenever we might get there.) The choice of twisted pair

had an impact on the noise immunity of the network and of course distance (as well as environment) in which the cable that is run affects the maximum distance the cable can be installed while still insuring a measurable signal.

Experience has also shown the biggest factor in reliability of a fieldbus system is the installation practices, simple things like making sure the connections have the proper torque, proper grounding practices, spacing between high voltage AC conductors and signal cables, and of course remembering fieldbus signals are wired in parallel so a short in one device can potentially short the entire segment unless short circuit protection is included in the field device coupler.

Culture is often less of an issue with these installations since most facilities deploying Fieldbus either migrated from pneumatic or "dumb" analog devices so the change to this new technology also brings with it the expectation of other changes to the way work is done.

By definition, field level networks include the communications between the field devices and their associated I/O card. Therefore, though it is still evolving, wireless networks such as ISA100, WirelessHART, and related industrial "personal area networks" with typical ranges of <100 meters should also be considered part of this level of the enterprise infrastructure foundation.

	Manchester Encoded Cable	Cables with and without Shield: 60VDC or 25VAC and < 400VAC	Cables with and without Shield: > 400VAC	Subject to any electro-magnetic exposure
Manchester Encoded Cable		10 cm	20 cm	50 cm
Cables with and without Shield: 60VDC or 25VAC and < 400VAC	4 inches		10 cm	50 cm
Cables with and without Shield: > 400VAC	8 inches	4 inches		50 cm
Subject to any electromagnetic exposure	20 inches	20 inches	20 inches	

Minimum separation distances between cabling (conversion rounded up for simplicity)

Controller networks+

Though proprietary networks still exist at this layer, they are predominantly now being based on Ethernet as the associated physical layer. Practically all buses have an Ethernet version where the protocol is bundled in an Ethernet package/packet. This is because the lower levels of the OSI model provide the physical and data link layers in which the data (application and/or user layer) is then carried. The information in the data packet/user message is packaged inside the complete Ethernet packet as it passes from the user layer, where the message is created, down to the physical layer, where it becomes either a voltage (cable) or frequency (wireless) so the 1s and 0s can be transferred from one location to another. When the message is received at the other end, the process is reversed, and the voltage/frequency is converted back to the message at the user layer of the recipient. The whole process is similar to sending and receiving a letter; the message does not care if it goes by foot, truck, ship, or plane as long as it eventually arrives and can be read by the intended person.

It is also because of this functionality that the control system supplier proprietary protocol as well as the various fieldbus protocols can run on an Ethernet backbone.

It is likely a matter of time before Ethernet-based field devices become more common, especially as Power Over Ethernet can provide signal and power via a single cable. However, the limitations for Ethernet continue to be distance (max. 100 meters) and a killer application where the high bandwidth (data) available with Ethernet is required.

At the Historian+ levels, and certainly at the interfaces between each layer with Ethernet, we need to be aware of security requirements and associated separation of systems. ISA99 standards propose several best industry practices, the key of

Wireless field level networks

Wireless is the new field level network that has the potential to open a range of process monitoring functions and a potential abundance of new applications once this data become available. We need only look at what we now do with our mobile phones to get an inkling of the possibilities.

However, like most industrial products, the challenge will be one of being able to take advantage of economies of scale. There are two impediments to making the economies of scale a reality in the wireless space. One is beyond our control, and that is it is unlikely a single wireless standard will be correct for all the different vertical segments/industries in which automation and control is used. Most notable of these is the high speed/low data packet size (discrete status bits) requirements of factory automation versus the lower speed/high data packet (analog information) needs for process automation. The second challenge is partially a result of the first, and that is the need for standards and like the fieldbus standard the resultant multiple versions of wireless networks standards to meet each niche.

Unfortunately, it looks like the process automation industry is compounding the problem with a potential three-way offering for a wireless standard being considered for submission to the IEC. The three standards include: ISA100, WirelessHART, and a "made in China" standard, all of which will likely be put forward for consideration, and as indicated above, the precedent has been set, so do not be surprised if the result is at least two process automation standards. Consequently, neither manufacturers nor end users will get the benefits of the economies of scale that might have been possible with a single standard, and everyone loses in the end.



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DD language continues to evolve

The Field Device Integration (FDI) project represents the next evolutionary step in Device Description language on which the three predominant field device protocols—HART, Foundation Fieldbus, and Profibus PA—are based. Consequently, FDI will have a significant impact on the future look and feel of digital field sensors, especially after the recent announcement that host suppliers ABB, Emerson, Endress+Hauser, Honeywell, Invensys, Siemens, and Yokogawa have joined the FDT Group, Fieldbus Foundation, HART Communications Foundation, OPC Foundation, and PROFIBUS Nutzerorganisation in pushing not only the development of this new standard but also incorporating it in their product offerings.

So what is FDI? When complete, FDI will be the replacement for all EDDL (IEC 61804 -3) based languages—HART, Foundation Fieldbus, and Profibus PA.

While EDDL is a common, text based description of a device, the text description is normally converted to a "binary DD" though a tokenizer before being shipped with the device. The above manufacturing company members of FDI have made it a high priority to harmonize the binary DD through secondary standards and tools, so the result will be a single binary format file regardless of the protocol of the device.

The EDDL file for each protocol will be processed through a tokenizer much like today—this also ensures backwards compatibility. Because each protocol is not exactly the same, but rather closer to 90% the same, it will be necessary to develop an FDI Developer environment for each of the three EDDL based protocols to assist them in defining how to map the various parameters of each protocol to the appropriate FDI parameters.

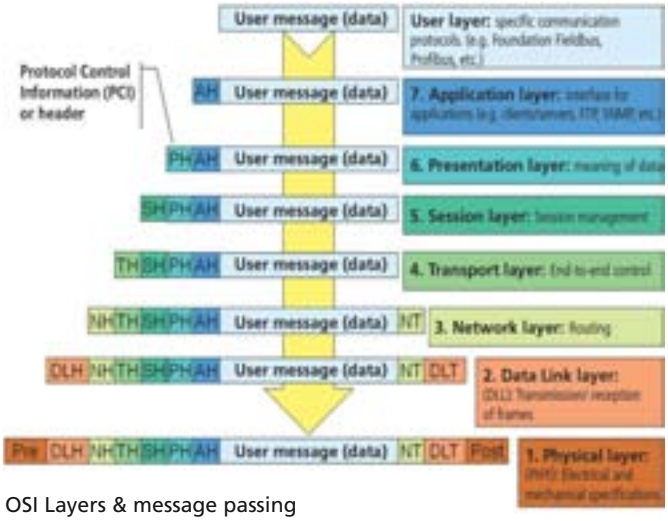
When complete, FDI will be the replacement for all EDDL based languages—HART, Foundation Fieldbus, and Profibus PA.

The resulting binary file from the tokenizer is then passed to a "packager" where it will be converted to an FDI file.

What is important to end users will be the interoperability of these devices, and that will be insured through the appropriately colored green "test tool" box, which will provide the necessary check mark from the appropriate organization that the devices are not only compliant with FDI but also backwards compatible with existing equipment.

Lastly, when the device is connected and communicating on the network, the process needs to be reversed with the DCS/host converting the FDI information into a format useable by the internal system databases. This is not any different than is done today, where each system needs to interpret the information from the field to the appropriate database register within the host.





OSI Layers & message passing

which is the concept of defense in depth (lots of speed bumps) and segregation of systems into cells, so should one cell become infected, it is not propagated to other parts of the system. A definite Demilitarized Zone (DMZ) between the business and control environment is also a must. In fact, one company insured quick separation between the DMZ and control system with a red colored patch cable to the related switches, so if necessary, they could quickly unplug this single connection and get physical separation of the system. Electrons have a hard time jumping an air gap. Remember process historians are designed to fill in missing data when they are reconnected to the control system that has its own short-term (typically 1 week) history buffer so a lost connection is not as onerous as it may first appear.

Field level networks are often taken for granted, however, as just shown, they should not simply be taken for granted because they are not as glamorous as other parts of the control system—if they do not work properly, the entire control system is susceptible to failure or as least wobbly results and that could easily lead to larger problems.

ABOUT THE AUTHOR

Ian Verhappen, P.E. (iverhappen@gmail.com) is an ISA Fellow, ISA Certified Automation Professional, and recognized authority on Foundation Fieldbus and industrial communications technologies. Verhappen operates a global consultancy Industrial Automation Networks Inc., specializing in field level industrial communications, process analytics, and heavy oil/oil sands automation.

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Fieldbuses for Process Control: Engineering, Operation, and Maintenance
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Opportunity for valve innovations

Despite economic downturn, opportunities abound for control valve innovation

By Hans D. Baumann

The control valve industry, being closely aligned with process control instrumentation, suffers similarly from the current recession. This drastic decrease in order came as a severe blow, following the hay-days in 2008 after the oil boom. Unfortunately, the future outlook is still bleak. The prospect of future high oil prices is expected to materialize only at the end of the current recession. Even then, spending is only expected on the production (upstream) side with refinery construction tabled due to lower gasoline demand and use of smaller cars.

This somber assessment is reflected in the last survey statistic published by *Valve Magazine*, where at the end of 2009, only 13% predicted an increase in business among their members, while 56% expected a decrease in shipments during 2010.

A somewhat brighter future is offered by the coming renaissance of atomic power plants. Those are the only, large scale, viable alternatives to all other "green" energy providing schemes.

Another avenue towards an increase in business is to export, especially into the developing markets. However, exports require competitive products and a strong manufacturing base. Unfortunately, the latter has suffered due to the past outsourcing boom. This has led to a dramatic shrinkage of manpower employed in the industrial sector. The percentage of workers in manufacturing as percentage of total employment went from 26% in 1965 to less than 8% today. As a result, we now have to import about 40% of all manufactured goods from abroad, mostly from China. This does not come for free, and we have to pay for it in U.S. Treasury Bonds to the tune of $650 billion in 2008 alone. This (current



account) deficit, which started in the 1980s, has reached a total of about $6 trillion, about one half of our total national debt.

There is a direct correlation between decrease in industrial manpower and our foreign debt (current account deficit). While manufacturing numbers started to decrease by about 20% between 1960 and 1980, due to the effects of automation, 1980 saw the beginning of outsourcing on a grand scale and with it the accelerated loss of manufacturing jobs. Analyzing foreign countries found a number of 16% of factory employment is needed for a country to have to have a positive trade balance. We reached that point in about 1990. At that time, our trade imbalance was only around $80 billion, mostly due to importation of foreign oil. In a recent interview, Jeffrey Immelt, the chief executive officer of General Electric Co., seemed to agree by suggesting increasing our manufacturing base to 20% of the total employed. That would be a great goal.

What can we do about this problem? We urgently have to reverse the trend and start rebuilding our manufacturing basis. Only by selling manufactured goods, control valves being part of it, can we increase our exports and in turn earn foreign currency to pay our debt. President Barack Obama was only half right when he called for a doubling of our export business. This can not be done without an increase in our current weak manufacturing base.

Here is where the government can help:

1. Give tax advantages to U.S. manufacturers.
2. Feed stimulus money into the manufacturing sector.
3. Provide low-cost loans to build new factories.
4. Encourage R&D efforts to make our products more desirable.
5. Allow to let the U.S. dollar to devaluate further. This will increase the cost of imported foreign goods (to be then replaced by local products) and decrease the price of U.S. exported items in order to be more competitive.

State of the art

At least 80% of all control valve designs originated in the 1960s. This to me is regrettable, since it represents a lack of progress and entrepreneurship.

While there was a major leap forward in the design of valve positioners during the 1980s and 1990s, where technologies jumped from analog to digital signals and circuitries enabled valve maintenance and even offered control functions, not much progress has been seen since. Yet, control valves still are a vital part of our control loops and, despite many predictions, have not been substantially replaced by speed-controlled pumps and the like. Part of the lack of attention for this vital part and its function as the final element in our fluid controlling loops is our seeming obsession with everything electronic.

One can also observe the basic hardware functions of a valve have not changed, leaving few choices for innovation. After all, we have not found a better way to control the rate of fluid flow in a pipe (the basis of all control modes, be it for pressure, temperature, or level). We still do it by creating a pressure differential, which then creates velocity. This in turn is converted by the valve plug or vane into turbulence (and heat), leading to a change in the amount of fluid passing a valve. This is a process called "throttling."

However, relying too much on old, established hardware can bring major disadvantages. The primary problem is old technologies offer no patent protection and can easily be copied. This can happen domestically by low-cost domestic repair facilities or overseas by factories in emerging countries, aided by lower labor costs. Sadly, the latter is unintentionally aided when U.S. manufacturers have valves made in foreign countries and in the process export vital know-how.

Why don't we see more R&D activities in the valve industry? My opinion is all new products are associated with risk, be it customer acceptance or worrying about performance problems. Corporate management, especially with a tight budget, tends to be highly risk-aversive. A second reason has to do with the way R&D activity is conducted—mostly on a computer. This restricts free and independent thinking, the basis of creativity, and limits "hands-on" experience. After all, there is no software as yet telling you how to invent something.

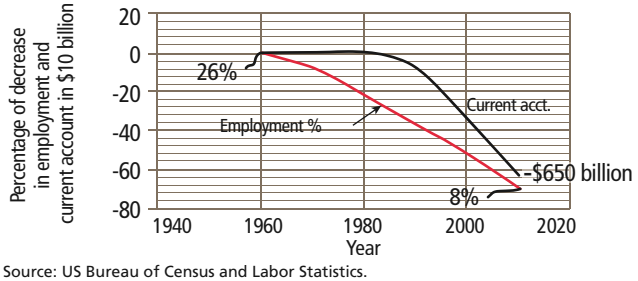
What is new

Yet, there are some efforts, albeit on a smaller scale, to add to the "state of the art" in control valve

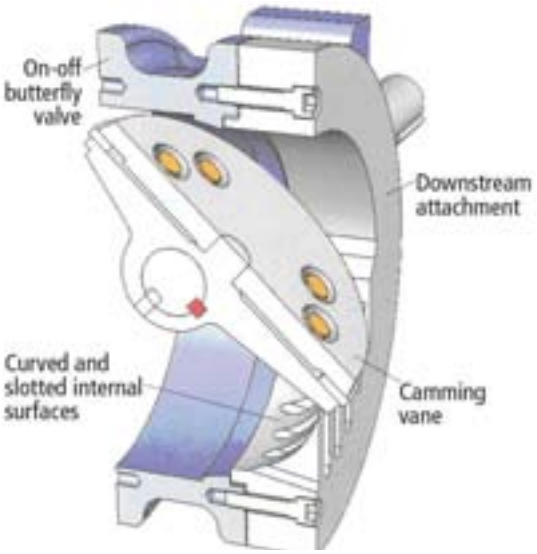
FAST FORWARD

- At least 80% of all control valve designs originated in the 1960s.
- Standards activities for control valves have taken a hit as a result of the severe economic downturn.
- A modified triple-eccentric butterfly valve is one example of improving control valve design.

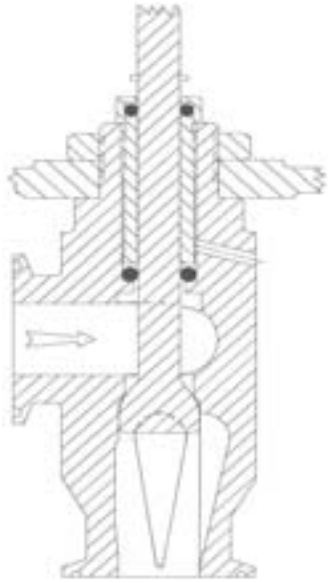
Relationship between manufacturing employment and current accounts



Source: US Bureau of Census and Labor Statistics.



A modified triple-eccentric butterfly valve
Source: YEARY CONTROLS, Chicago, Ill. (Patent applied for)



Installing a plug from below creates a "negative feedback" situation and assures dynamic stability.
Source: Spence Engineering Company, Inc.

design. One example is a modified triple-eccentric butterfly valve. Here, is a way to convert a standard, commercial, on-off butterfly valve into a well-performing control valve, by adding a downstream attachment having curved and slotted internal surfaces providing, in conjunction with the camming vane, an equal-percentage flow characteristic and, at the same time, offer low noise and anti-cavitation features. Thus, by combining the advantages of a low-cost, tight shut-off, and the high pressure capabilities offered by such a valve with a replaceable static throttling device, one can offer a better substitute for many standard globe and rotary control valves.

The slotted areas between the teeth are gradually opened by the lower half of the vane. Such reduced flow areas increase turbulent frequencies of passing gaseous fluids. This leads to a substantial attenuation by the downstream pipe wall, hence a lower aerodynamic noise level. A similar effect is achieved with liquids. Here the "coefficient of incipient cavitation" (the pressure ratio signaling the onset of cavitation) is increased, allowing for higher pressure drops. However, even if cavitation should occur, it is only a local phenomena, restricted to near the outlet of the slots, thereby avoiding the pipe-damaging "super cavitation" typical with standard valves discharging directly and unimpeded into a piping system.

Another example is a new control valve design especially developed for the bioprocessing industry. The design challenge is such a valve has to be aseptic, to be self-draining, have a good flow characteristic, and be dynamically stable. Highly polished angle valves meet most of those requirements, since they can drain directly into the top of a vessel, with flow entering from the horizontal port. The problem with such designs is the valve plug is inserted from the top, which means the plug is closing down against the seat where the fluid pressure tends to force the plug down and close the valve. This creates a "positive feedback" force and can lead to dynamic instability, even slamming against the seat. This is especially problematic with larger valve sizes.

One valve design overcomes such problems by installing the plug from below, i.e., pulling the plug up against the seat (and the inlet pressure). This creates a "negative feedback" situation and assures dynamic stability. Another feature of this design is the use of throttling channels in the sides of the outlet ports, where they are easy to clean and polish and where the circular surface areas between such "slots" provide ample guides for the moving valve plug. Removing the throttling surfaces from the plug, as was customary, and placing them inside the slots avoids other potential problems such as unsteady flow caused by wall attachments and the tendencies to cavitate.

A variation of the flow capacity (Cv) of such a valve can easily be achieved by simply altering the number of parallel slots. Finally, by inserting the plug from below, one can dispense of a removable bonnet at the top of the valve in order to reduce cost and eliminate a potential leak source.

Standards activities

Standards activities for control valves are in the realm of ISA75. Here too, the severe economic downturn has left its mark, and some standards activities are now relegated to the Working Group 9 under International Electrical Commission (IEC), Switzerland, Committee 65B. But even here, activities are relegated to updating existing valve and positioner standards. The most important revision is on IEC 60534-8-3, "Control Valve Aerodynamic Noise Prediction." The new draft, currently in preparation, departs from the current "science-fundamentals" based model to one heavily inspired by empirical data, which makes the document less "vendor neutral" and adds substantial mathematical complexity. Standard 6053-2-1 on control valve sizing also gets updated. There are no basic changes planned in customary sizing equations. What is proposed is to simplify some equations and make them more user-friendly.

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Implementing MES boosts profits



Replacing existing custom-built production information improves operations

By Bianca Scholten

As a consultant, I visit many production facilities. Currently, several factories are replacing their 10- to 15-year-old, custom built manufacturing execution systems. An even larger amount of industrial companies have only just become aware of something called MES. They still have to convince management that MES is worth the investment.

What is MES? MOM?

What is MES? That is the first thing to explain to the board. And what is the relationship between MES and other terms like manufacturing operations management (MOM)? Why do companies invest in these kinds of information systems? And why should you replace the old system, to which people are so attached?

"The term *MES* arose in 1991," said Jan Snoeij, board member of MESA International. "Multiple people claim they invented it. At that time, it stood for manufacturing execution systems. Today, it means manufacturing enterprise solutions. After all, MES is more than just a system for production control. Issues such as quality, inventory, maintenance, product data management, and product life-cycle management can't be viewed as separate from the MES domain. ... That's why we changed the term in 2004."

In one of its white papers, MESA International distinguished 11 manufacturing execution activities, which later gained recognition primarily thanks to the MESA honeycomb model.

Simply put, the original concept, "Manufacturing Execution System," concerns information systems that support the things a production department must do in order to:

- Prepare and manage work instructions
- Schedule production activities
- Monitor correct execution of the production process
- Gather and analyze information about the production process and the product, and

feed this back to other departments, such as accounting and logistics
- Solve problems and optimize procedures

MOM concerns the activities within the production department, the maintenance department, the lab, and the warehouse. MES systems mainly focus on a part of MOM, namely the support of activities within the production department. For the other parts of MOM, other kinds of information systems are available. The function of asset management systems focuses on maintenance departments; laboratory information systems (LIMS) support the activities within the lab; and warehouse management systems support warehouse activities.

Advantages of MES

At MESA's European conference in 2007, the organization asked the audience several questions. They handed out green, red, and yellow cards that stood for the answers "Agree," "Disagree," and "No opinion," respectively. One of the questions was, "Who believes that MES can produce significant advantages for industrial companies?" I was in the audience and thought to myself, "Hmmm. Who believes? ... Apparently, MES is a belief."

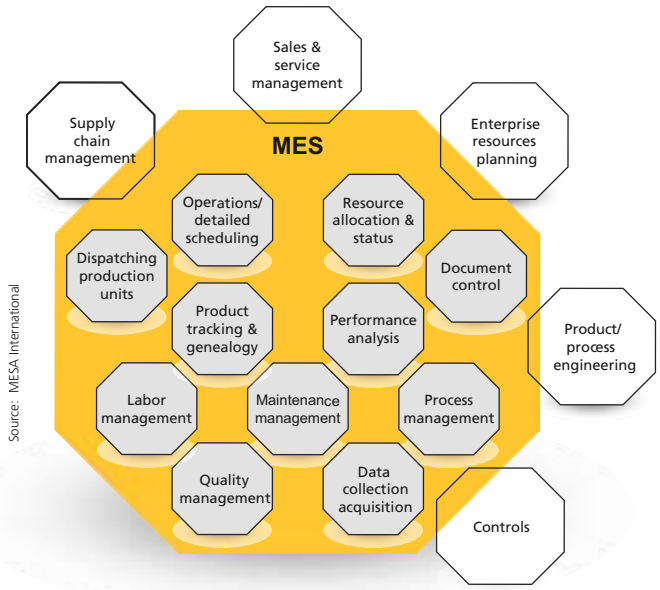
By the way, most of the audience held up a green card. Believing is not the problem. But wouldn't it be great if plant managers and IT managers could walk up to their bosses with hard-and-fast numbers?

Fortunately, market research agencies have not let us down. For example, at its members' request, MESA commissioned a study into the way in which manufacturing companies improve their financial performance, and how they justify their investments in production automation software. MESA hired a consulting firm to create an analysis program. The analysis team's Internet questionnaire produced 151 valid responses.

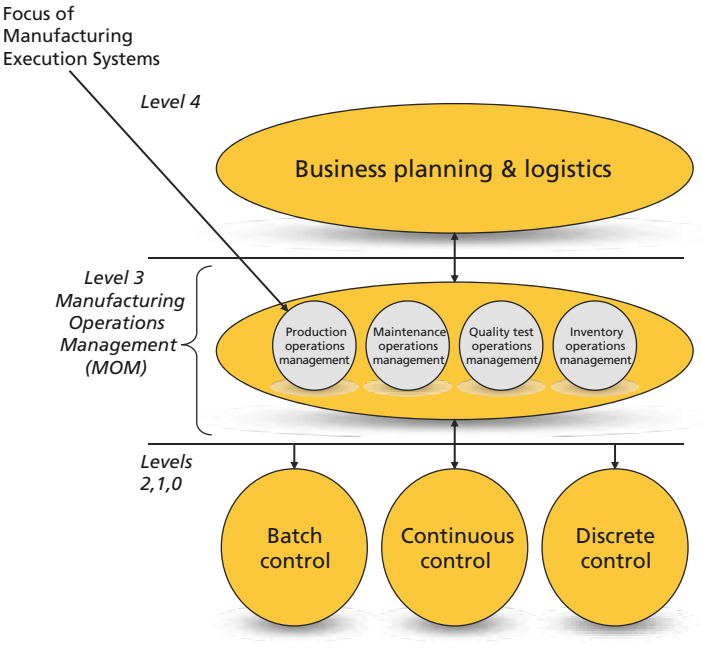
Based on a list of widely used key performance indicators, like labor cost per unit and on time delivery, respondents to the study indicated how many improvements they had realized in the past three years. The consulting team then divided the companies into two groups, "Business Movers" and "Others." The Business Movers are those companies that demonstrated considerable improvements, in breadth or in depth. That is, they started performing more than 1% better on six of the 11 business metrics in the study, or they demonstrated more than 10% improvement on at least one of the business metrics. These Business Movers serve as a model for other companies. What qualities did they have that were possibly the source for realizing so many improvements?

The report describes the profile for these companies as

FAST FORWARD
- MES systems mainly focus on a part of the manufacturing operations management, namely the support of activities within the production department.
- The faster the company can feed results back to employees, the faster employees can take corrective action.
- Implementing a workflow management system could lead to shorter turnaround times and improved delivery reliability.



Manufacturing execution activities in the honeycomb model.



The relationship between MES and MOM: MES and MOM belong to level 3, which is the level below the enterprise resource planning systems and above the process control systems.

follows: They are fast; they have coupled their operational goals to their financial and business goals; they know their results; their plant activities are profitable; they concentrate on what is important; they use software applications; and they have, in general, a return on investment of two years or less on their investments in plant software.

According to the report, speed is one of the characteristics of the Business Movers. The faster the company can feed results back to employees, the faster employees can take corrective action. Automated data collection can help make this information available earlier. Many Business Movers feed results back within 24 hours, or even in real time. They more frequently use automated data collection than do others.

Another conclusion from the report is Business Movers more often use MES and information dashboards. This group clearly shows more improvements than do companies that have not adopted these types of systems.

MES is a belief, but various market research agencies think they have found a causal link between the use of MES functionalities and improvements to operational and company-wide performance. Keep in mind, however, these studies are not always conducted at the

The business case for MES at Palm Breweries

Before we started to use the new MES system, we had a custom built production information system that we had been using for about 10 years. For a long time, we were very happy with the old solution, but toward 2006, it was outdated. It did not cover all the modern requirements anymore. For example, the requirements for tracking and tracing had become much more rigid, and the old package did not support that. Furthermore, we of course wanted to continuously improve things, and therefore we need good information, e.g. about the usage of raw materials and consumables and quality costs. In the old system, it took us two or three days to collect the right data, so that was hard to do. We knew we were overseeing things and savings had to be possible in the process. But it was not feasible to adjust the old system, which had become basically unreliable, because we had not updated the latest versions of the software platform.

Finally, we concluded we wanted to purchase a new system, which had to make it easier to comply with regulations. We also defined other advantages and tried to quantify them. We assumed the system was going to cost approximately $812,800 (€600,000), and for that, I had to present the business case to the Board of Directors.

We estimated the consumption of raw materials could be lowered 1% and the usage of consumables 2%. Up till then, we used to calculate our losses in Excel, by comparing the purchased amounts of raw materials with the amounts of finished products we had sold. But it was very cumbersome to analyze in which process step and during which week the exact loss had been generated. Now, with the new system, we know our losses per lot and per process step, and we can easily look up the lots of specific periods of time and compare them. The same is true for consumables. Thanks to the storage of historical data in a historian, we would gain better insight in the process, and we would be able to relate quality data to production data. This way you discover where savings are possible concerning consumables and energy.

We also made an interface between the planning functionality in MES and the level 4 purchasing system. By improved planning and less rush orders, we estimated we would be able to purchase packaging materials at a lower price.

Quality also was a very important driver. The MES system should be able to lower the costs of solving quality issues 25%. We made this estimation by listing historical incidents and understanding how the future MES could have avoided these issues. Furthermore, the risk of unavailability of the old software system would end.

We also saw some advantages that we could not quantify. For example, we have a quality index that tells us what percent we produce according to desired specifications. Our standard is 95%, but thanks to a better monitoring, 97% should be possible. We wanted to realize that by purchasing a laboratory information system but also by improving production tracking and tracing. That would make our quality even more stable. Quality is the trump of our brewery. It is possible other breweries purchase a MES mostly to lower costs, but for us, the focus was on quality.

We also fill and package beer for other breweries. One of the advantages that we could not quantify was the satisfaction of these business-to-business customers. For those clients, it is nice if they quickly and accurately receive reports about the products we made for them. Making those reports in the old situation took a long time, and it was error prone. Now, we have automated this process, and reports are sent in near time to our customers. We know from our own experience—we have our beer filled in cans by another brewer—that you trust in your supplier when they provide insight this quickly, as if it were packaged on our own site.

We also found the employee satisfaction an important driver. We want to be ahead in the market and offer a professional working environment. The old system led to frustrations, whereas the people are happy to work with the new system.

We have realized the most important goals. Some we even surpassed. On others, like the reduction of raw material losses, we are still working. Of course, it is difficult to prove specific savings were caused by the software package, but we believe in it. We earned back the investments. We had calculated a onetime savings of $135,000 (€100,000) and a yearly saving of $101,000 (€75,000). We saved two full-time equivalents, so there we already have our yearly calculated savings. We are satisfied. In the mean time, we keep discovering new possibilities in the system that is all extra benefit.

SOURCE: Alex De Smet, head of production, Palm Breweries (interviewed by Bianca Scholten in January 2010)

request of an independent organization, but often on behalf of vendors. In that case, you might question the market research agency's objectivity.

Ultimately, of course, we have to keep using our own sound judgment under all circumstances. Let's try to explain those expected improvements just by thinking logically.

Using the detailed production scheduling functionality of MES may possibly lead to higher efficiency. Such a module can quickly calculate diverse options (simulation) and, in so doing, propose the most efficient combination and sequencing of orders, changeovers, and cleanings.

If you implement a module for recipe management, assembly instructions, or standard operating procedures, you can synchronize master data automatically with the Bill of Materials in the enterprise resource planning system. Particularly for companies in which recipes often change, this delivers advantages. There is a smaller risk that operators will be using outdated instructions, and changes from R&D can be more quickly incorporated. Because the recipes are flexible, you can always use the least expensive raw materials, as is common practice in the feed industry. The software can also automatically send product-related parameters to the PLC-SCADA layer. Because operators no longer have to manually transfer those data, they win back time, and the risk of error is reduced.

A historian module can take over the operator's task of collecting various data, so they can concentrate on their real task of monitoring and guiding the process. Moreover, a historian gives you the opportunity to store large quantities of data over a long period of time in a central location. This improves access to historical data and lays the foundation for process analysis and optimization.

In many plants, people use paper forms and spreadsheet programs to create reports. For example, there are pharmaceutical companies that produce more pounds of paper batch reports than they do pounds of end product. Customers and government



require tracking and tracing, and as long as you are recording data, why not use all that information for other objectives, such as continuous improvement? But collecting data from paper forms and Excel files, converting units of measurement, and then making comparisons takes a lot of process engineers and production managers' time—time they would rather spend improving the process, quality, efficiency, and so on.

MES can provide more insight by quickly generating a variety of reports and charts based on the collected data, relating to quality, overall equipment efficiency, performance by shift, by day, by batch, and so on. These reports form an important input for team discussion in order to get everyone on the same wavelength and to concentrate on what is really important.

But reports alone will not get you there. They do offer a basis for making various comparisons, but by definition, they look at the past, and they always contain the same information. Reports are static. A company that runs its plant based on reports is like a car owner who drives while looking in the rear-view mirror. It is better to know what is happening now, to respond proactively, and to make adjustments. And because you cannot be everywhere at once or talk to everyone at once, it is useful to have a dashboard.

Finally, workflow management modules can function as the orchestra conductor. They streamline processes in which different departments are involved, by pointing out tasks and priorities to employees and by providing insight into who is working on which order and where. Implementing a workflow management system could thus lead to shorter turnaround times and improved delivery reliability.

Replacement of existing MES solutions

Those pioneers who already built their first custom MES 15 years ago are facing the challenge to replace the old system. In many cases, this is necessary because they are based on outdated technologies and maintenance costs are high. IT managers have nightmares because the plant has become completely dependent on the knowledge and skills of one technician. Extending the MES with new functionality is expensive or impossible. So from an IT cost efficiency point of view, the replacement of the old custom built system by a system with out-of-the-box functionality, which can be supported by different system integrators, has many advantages. But how is the IT guy going to convince the users who by now are so attached to this solution that was built with a complete focus on their specific requirements? A new, standard system will never have the same perfect fit. It is like the difference between haute couture and prêt a porter. You do not have your clothes made by a famous designer; you buy them at the mall. It is not a perfect fit, but you are okay with it because the price is acceptable. Now, convincing the users to move from designer solutions to standard, fit-all systems is the biggest challenge in replacement situations. You will have to carefully stitch in the change to avoid a culture shock.



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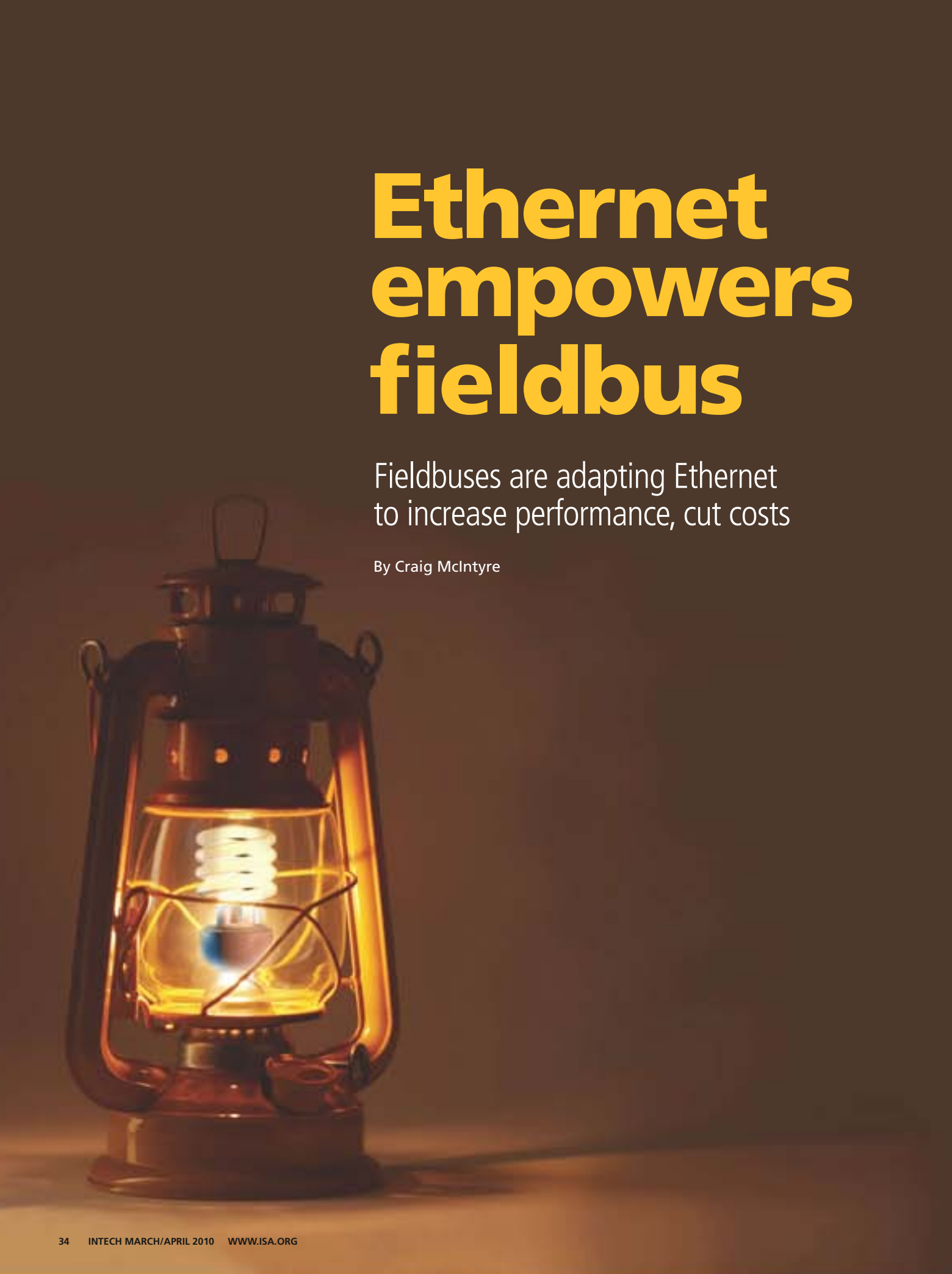
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Ethernet empowers fieldbus

Fieldbuses are adapting Ethernet to increase performance, cut costs

By Craig McIntyre



Fieldbus networks have been around for over a decade, delivering value in industrial automation applications worldwide. In most cases, a fieldbus network is used to link field devices to a host computing system in a process plant.

The field devices in question are most typically instruments, analyzers, and modulating control valves. The most popular types of instruments are flow, level, temperature, and pressure transmitters—although a typical process plant will also include many other types of process variable transmitters.

Common analyzer types measure the chemical composition of parameters such as moisture, carbon dioxide, methane, and other gasses and liquids. Modulating control valves provide continuous flow control of liquids and gases.

The host computing systems linked to field devices include control systems, Enterprise Resource Planning (ERP) systems, and asset management systems. The control systems in question are usually basic process control or regulatory control systems. Other control systems with fieldbus links to field devices include advanced process control systems and safety systems.

The benefits of linking field devices to host computing systems via a high speed data link like fieldbus include:

- Networked device configuration and health management saves money
- Networked device documentation saves money
- Predictive maintenance increases uptime
- Predictive maintenance improves performance
- Predictive maintenance cuts maintenance costs

In almost all cases, fieldbus is used to replace a hard-wired 4-20mA one-way connection from the field device to the host computing system. This 4-20mA signal was used to transmit the measured process variable to the host computing system.

In contrast, fieldbus provides a high-speed two-way data link that can transmit copious amounts of information between the field device and the host computing system. The most important bits of information are data related to instrument health, which can be used by the host computing system to schedule calibration as needed and for predictive maintenance.

Scheduling calibration only as needed instead of on a periodic basis saves money because field devices are not calibrated when operating within parameters. As needed, calibration improves performance because more frequent calibration can be performed on critical field devices that are drifting out of range.

FAST FORWARD

- Most fieldbus organizations are transitioning to an Ethernet-based protocol.
- Ethernet platforms are expected to provide a better fieldbus solution in terms of price/performance ratio.
- End users will welcome the change as it lowers their costs and simplifies installation and maintenance.

Predictive maintenance is perhaps the most important benefit delivered by a fieldbus system. Using the data delivered by fieldbus, a plant can predict problems before they occur. Maintenance can then be performed on a planned basis as opposed to a reactive basis, saving money and improving safety.

But the main benefit of predictive maintenance is avoidance of downtime. If a field device is found to be failing, it is often possible to repair or replace the device before it brings down the entire process.

Fieldbus is a generic term for a number of digital industrial networks, including but not limited to, Foundation Fieldbus, HART, EtherNet/IP, Modbus TCP, and Profibus. Many fieldbus networks, including those mentioned above, are transitioning to Ethernet-based protocols for a variety of reasons.

Why Ethernet?

Benefits of fieldbus have been well documented in many process plant applications for years, and most of these fieldbus installations use proprietary protocols as opposed to Ethernet-based protocols. But Ethernet-based fieldbuses can often provide better performance at a lower price, while also simplifying installation and maintenance.

When the term Ethernet is mentioned in the commercial world, it is usually in reference to an Ethernet network utilizing the TCP/IP protocol. In the industrial world, the term Ethernet only identifies the underlying hardware and not the protocol, which can be found in various flavors. The bad news is there are many competing industrial Ethernet-based protocols. The good news is they can all run on the same underlying Ethernet hardware, often simultaneously. (See sidebar for more information.)

An improved price/performance ratio is perhaps the main benefit of switching to Ethernet. A few years back, one of the leading fieldbus network organizations was trumpeting the fact that over 1 million fieldbus devices using their proprietary protocol had been sold over its

10-year life. This number may have been impressive compared to other proprietary competitors, but it is trivial in comparison to the number of Ethernet devices installed worldwide.

As the leading protocol for computing connectivity, there are billions of Ethernet nodes installed worldwide, mostly in commercial applications. Large numbers generate economies of scale, allowing providers of Ethernet hardware to continually drive down costs and increase performance. And because Ethernet is a worldwide standard, vendors compete based on price/performance ratios and thus have tremendous incentive to deliver the best bang for the buck.

Fieldbus organizations have taken advantage of commercial Ethernet economies of scale to deliver higher speeds at lower costs. Higher speeds allow quicker update times for monitoring applications. Depending on the characteristics of the process being controlled, higher speeds can also enable real-time control.

Because Ethernet is so pervasive, installation and maintenance are easier for a number of reasons. First, it is much easier to find technical personnel familiar with Ethernet as opposed to a proprietary network. Second, hardware and software tools for installation and troubleshooting are widely available at low cost. Finally, it is often possible to lean on corporate and plant IT personnel for support and maintenance because IT folks speak Ethernet. By converging their industrial and office networks, end users have fewer variant networks to maintain and gain more leverage when integrating technologies and communications.

Ethernet provides concrete benefits to fieldbus, and many users are taking advantage in a variety of applications. In particular, Ethernet connectivity is bringing fieldbus to the Programmable Automation Controller (PAC) level.

Fieldbus Ethernet enables PACs

In early implementations, most fieldbus control systems were of the DCS variety. Now that Ethernet is providing a common communications protocol, fieldbus is becoming a viable option for processes controlled by PACs. This can decrease costs for process control systems as a PAC is typically less expensive than a DCS.

A leading manufacturer of floor coverings for commercial and residential applications uses a combination of PACs and EtherNet/IP-enabled field devices to improve product quality, promote green manufacturing methods, and enhance production efficiencies. At one of their plants, this manufacturer uses a PAC as their main real-time control platform. The PAC comes with EtherNet/IP built in, so the plant would like to use this protocol as their Ethernet-enabled fieldbus.

Consequently, an EtherNet/IP-enabled Coriolis flowmeter was required for measurement and control of mass flow in one of the plant's continuous processes related to colorant control. The mass flowmeter selected was found to have superior accuracy in competitive field trials conducted by the manufacturer. Direct connectivity from the meter to the PAC via EtherNet/IP provides a number of benefits.

The meter is capable of simultane-

Direct is best

When flying from one city to another, direct flights are always better than those requiring an interposing connection at a hub airport. It is much the same with fieldbus connections to higher level host computing systems, where direct connection from the field device to an ERP or asset management system is preferred to a connection via an interposing control system.

It is no longer correct to assume every field device must be connected to a control system to have value.

Bypassing the control system and going directly to a host computing system like an ERP or asset management system is made easier by Ethernet-enabled fieldbus. That is because most every host computing system is capable of Ethernet communications.

In a typical process plant, most field devices are not directly associated with real-time control and are instead used primarily for monitoring. Some facilities have found up to 70% of their field devices do not have any associated control functions. It is no longer correct to assume every field device must be connected to a control system to have value.

For example, environmental applications such as EPA monitoring and reporting require multiple field devices that do not need to be connected to the control system. ERP applications such as inventory management need field device input, but do not require the millisecond update speeds or deterministic behavior associated with control systems.

Bypassing the control system provides a number of benefits. First, a direct connection eliminates the need for intermediate hardware components and software systems. Second, the field device in question may not need to be part of the control system's overall validation and maintenance program. Third, field device access can be controlled through existing IT security systems.

Information from these non-critical field devices can be conveyed directly to process monitoring applications via standard Ethernet networks and wireless field gateways. Not only is the primary information delivered in fully defined engineering units, but device status can be continually monitored and communicated on an event driven or periodic basis.

Remote servicing tools and asset management applications working at the network level can also configure and manage connected devices. Standard IT security and data management tools can be used to control access.



Field devices like these flowmeters deliver more value when connected to control systems, asset management systems, and ERP systems via an Ethernet-enabled fieldbus.

ously measuring multiple parameters including mass flow, product density, process temperature, volume flow, custom concentration, and viscosity. The plant wanted the ability to monitor these parameters without having to run multiple wires, and the solution was the high-speed 100 Mbps EtherNet/IP protocol.

The meter's advanced diagnostics parameter monitoring can now be used by the plant to predict process influences from coating, buildup of solids, corrosion, erosion, and entrained gas conditions. Predicting problems before they occur enables predictive maintenance, cutting costs, and reducing downtime.

Further benefits identified in this application included a 40% reduction in device commissioning time and a 25% reduction in loop identification, device integration, and process loop tuning time. Immediate recognition of the meter as a network node is another benefit, along with transparency of the meter from the factory floor to the enterprise system.

There is no doubt that fieldbuses will continue to evolve towards Ethernet-based implementations. To remain competitive, the fieldbus organizations will have to take advantage of Ethernet's superior price/performance ratio.

End users will demand Ethernet-enabled fieldbuses to simplify direct connections from field devices to host computing platforms such as ERP and asset management systems. For the growing number of process plants using PACs, Ethernet-enabled field buses will be seen as a natural fit.

More and more plant floor technical personnel will become familiar with Ethernet, encouraging its use in process plants. Internal IT personnel will also prefer Ethernet-enabled fieldbuses because of similarities with corporate computing system networks. The Ethernet-enabled fieldbus bandwagon is rolling forward, and suppliers and end users are jumping on.

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All fieldbus Ethernets are not created equal

Fieldbus organizations like to boast that their particular flavor of Ethernet network is the best, adhering most closely to commercial Ethernet TCP/IP implementations while still delivering the real-time performance and reliability needed for industrial applications. While those claims are for end users and system integrators to judge, there is no doubt Ethernet-based fieldbuses come in many different and often incompatible varieties.

For example, EtherNet/IP and PROFINET are Ethernet-based fieldbuses, but it is not possible to mix and match components adhering to these two standards without some kind of gateway and/or translator. In other words, a PROFINET-enabled field device cannot be directly connected to a Programmable Automation Controller (PAC) with an EtherNet/IP port.

One solution to this problem is to only use field devices and controllers compatible with one fieldbus. Unfortunately, this is often not a viable option as most process plants have existing field devices adhering to different fieldbus standards.

Even for a greenfield plant or process, it is usually not possible to purchase all of the required field devices from vendors adhering to one Ethernet-enabled fieldbus standard. Certain specialized instruments, analyzers, and control valves are often needed, and these field devices need to be connected to the control system and fieldbus of choice.

To cope with this issue, many vendors make gateway devices that convert one Ethernet-based fieldbus protocol to another. For example, a gateway can allow connection from a Modbus TCP-enabled field device to a PAC with EtherNet/IP connectivity.

Although gateways solve the incompatibility issue, it is best to minimize use for a number of reasons. First, gateways add to the overall cost of the fieldbus installation. Second, gateways add complexity in design and maintenance. Finally, gateways require additions to maintenance inventory and increase stocking requirements.

There was hope Foundation Fieldbus High Speed Ethernet would become the industry standard, but not all DCS vendors subscribed to this effort. End users long for the day when one Ethernet-enabled fieldbus emerges to rule them all, but until that time, it is best to use compatible components to the greatest extent possible, with gateways accommodating outliers.

Industrial Ethernet all the rage



Industrial Ethernet is designed to deal with harsh environments, data collisions, factory noise, factory process needs

By John Rinaldi

Ethernet is a well known and recognized technology in the home and office environment. Recently, it has become the hottest trend in moving data in industrial applications on the factory floor. The factory floor, however, is a much different environment than home and office environments. This article highlights the differences between industrial and commercial Ethernet by comparing communication needs, process concerns, environmental challenges, and hardware.

Let's communicate

Industrial Ethernet has unique requirements based on two-way communications. To understand this, think about a possible application at a bottle-filling plant. Assume the plant is creating a new microbrew beer, Automation Ale. The filling operation will be run by an industrial Ethernet network.

The network works well because it uses "handshaking" to ensure message delivery. To illustrate this attribute, let's say our bottling device begins filling a bottle of Automation Ale at the command of the controlling PLC. The PLC is also responsible for sending the "stop filling" command when the bottle is full. If the message is lost on the network, the PLC is aware because it does not receive a delivery response, (part of the handshaking) so it knows to resend the command.

In the office setting, such a lost transmission is rarely important. If a web page gets lost in transmission, the user simply presses "refresh." In the production setting, though, we cannot wait for Automation Ale to spill on the floor before someone manually turns off the filler. The handshake saves ale, money, and time.

In an industrial Ethernet network, we also incor-

porate collision detection. If two messages collide in our network, the controlling PLC can resend the message to the device until it receives a delivery notice for the device. Ale continues its controlled pour, and no one is crying over wasted beer.

Automation Ale is quite popular in our scenario, so assume we need a few dozen bottlers, valves, sensors, and a PLC in our network. The operation must run at peak efficiency; an office Ethernet network would not accomplish this goal. That is because there is no collision detection.

Other factors to consider

Operations concerns: An area of concern regards the cost of downtime. When a network goes down in your office setting, it is an inconvenience, and some work may be impossible. Often though, an employee will simply need to move on to another task and tackle it without use of the Internet.



In a production setting, that downtime is costly. Assembly lines operating with continual processes can be rendered nonfunctional if one aspect fails. Critical processes could be ruined, leading to lost material and money.

Think for a minute of a factory producing tempered glass for windows. A continuous flow of glass moves from pour, to cut, over an assembly line a mile long. The glass flow progresses through specific heat-ups, cool-downs, and rests to properly temper it to meet production specifications. If the line seized, the factory would be left with a mile of scrap glass. Much of it that would need to be removed manually due to the fact it had cooled hard on a portion of the line that was meant to deal with hot malleable glass.

When designing an industrial Ethernet network, you must consider options that make your network reliable. That often leads to increased costs.

Security: In an office setting, the information traveling through the network can be confidential and important, thus an office Ethernet network must guard against unauthorized use. The same is true in an industrial application. Another security threat in the industrial setting is the risk that an employee may break the system accidentally, creating a Garbage In/Garbage Out scenario or bringing the device or network to a complete halt.

FAST FORWARD

- An industrial Ethernet network needs to incorporate collision detection.
- When designing an industrial Ethernet network, consider options that make your network reliable.
- Industrial Ethernet topology options include: star, tree, line, and ring topologies.

Office and shop floor differences: You would not expect to see someone in the industrial setting wearing Italian suits or expensive leather shoes because it is much more suitable for them to be wearing blue jeans and steel-toed boots. These choices offer more protection from the environmental factors in the factory. The same attire considerations need to be taken for your Ethernet networks. Industrial Ethernet cables, switches, and connectors need to withstand the unique and harsh criteria in an industrial setting.

Environmental concerns

Temperature: Heat and cold are two factors that can have a major effect on a network. Cold is particularly damaging. At relatively cold levels, near freezing, a cable is susceptible to impact, which can cause a break in the cable, destruction of the protective jacket, or attenuation. At even colder temperatures, the cable may become brittle and break through no large force, but instead through simple bending.

Heat is also damaging. The protective jacket may melt, leading to shorts and vulnerability. Heat also causes attenuation over time.

Chemicals: Chemicals may cause a jacket to dissolve or change shape, leading to a shorter life and worse performance. Some solvents can also directly impact the internal cable should the protective jacket not be effective. Radiation, especially UV Radiation from sunlight, can cause discoloration and degradation of the jacket. Humidity can also degrade the cable.

The industrial Ethernet environment is harsh, and office Ethernet applications were not created for such environments. Taking measures to physically protect cables and connectors can minimize, or even negate, the effects of an industrial environment.

Factory noise: Electric and magnetic noise generated by large motors and high voltage devices can distort data transfers on the network.

Vibrations: Some processes may create vibration, which can cause degradation of the jacket

and disconnection if poor connectors are used. You must consider what will happen when the machines switch is turned on.

Other notable differences

Topology: Commercial Ethernet is almost always configured in a star topology. Industrial Ethernet has many different topology options to fit diverse industrial applications. The topologies include star, tree, line, and ring topologies.

Heavy and light duty: Office Ethernet components are designed for a base level of use. Industrial Ethernet components can be considered for multiple levels of use. Thus, industrial Ethernet components can be divided into heavy and light-duty categories.

Cable: Cables can be classified as heavy or light duty. A light-duty industrial Ethernet cable may have slightly higher quality jacketing than office Ethernet cable. The cable may even be an office Ethernet cable if the conditions do not require extra protection. As you rise to heavy-duty cable, though, the jacket and metals improve. At some point, you begin to see complex and thick jackets around incredibly high-quality cable. Heavy-duty cable is more expensive than light-duty cable, so it is only used when necessary.

Connectors: Connectors can fall on a spectrum from office to light duty and up to heavy duty. Typically, industrial Ethernet connectors will not rely on basic snap-in lock mechanisms on the same level as office Ethernet. Instead, heavier lock mechanisms are used. In heavy-duty applications, sealed connectors are often used.

Industrial light and heavy-duty parts carry a premium price tag when compared to commercial components.

Ethernet is quickly becoming a well known and used technology on the factory floor. It offers cost, data volume, and transmission speed improvements over its fieldbus predecessors in industrial applications. Industrial Ethernet is able to effectively deal with harsh environments, data collisions, factory noise, and factory process needs. It is still Ethernet, just Ethernet designed to fulfill unique industrial needs.

ABOUT THE AUTHOR

John Rinaldi (jrinaldi@rtaautomation.com) has a great deal of experience in industrial control and is the coauthor of the book *Industrial Ethernet*. His company Real Time Automation, Inc. specializes in industrial networking software stacks, OEM modules, custom design, and off-the-shelf gateways to bridge protocols (www.rtaautomation.com).

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Industrial Ethernet, 2nd Edition
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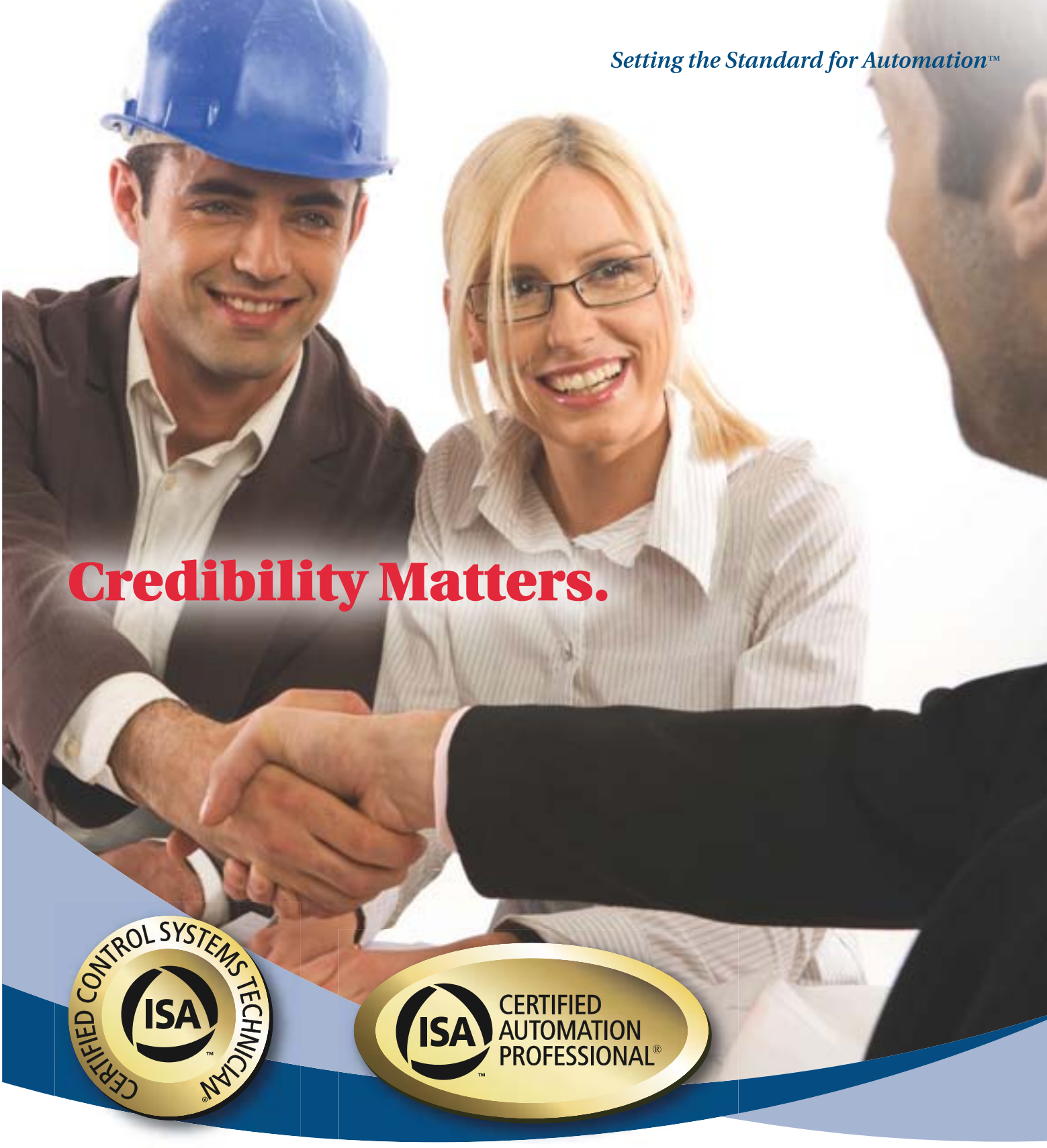
Fieldbus Foundation's High Speed Ethernet
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The '*emerged*' skill crisis…

By Dr. Ken Ryan



Editor's Note: Dr. Ken Ryan is an education professional passionate about giving people the education and know-how to improve manufacturing comments on the issues in North America. This is one perspective, and InTech *is interested in perspectives from other parts of the world with similar issues.*

We used to speak about an "emerging" skill shortage in this country as much as we used to talk about the "potential" financial crisis.

Well, they both happened!

Not only is the skill shortage knocking on our front doors, it is now residing in our living rooms as we permanently rearrange our furniture.

There are two principle reasons for this:

1. We have duped ourselves into believing we can build a sustainable economy without the durable manufacturing activities that characterize those nations threatening to eclipse us.
2. We face not only an aging of the skilled workforce but a collateral erosion of the education assets required to replenish the supply.

Reversal of the first problem demands a steeling of the collective social will that may be beyond the American public's attention span. In this case, "Resistance is futile!" We may as well sit down, collect our government checks, and wait for the end; however, I believe it is still (barely) within our power to reestablish the preeminence of manufacturing in our society. Given this resolve, we must point out why we are in this predicament and then focus on solving problem number two.

First, in a self-absorbed focus on academic purity, pensions, and seniority, we educators have participated in the isolation and politicization of the American education system and taken our collective eye off the prize of service to the next generation.

Next, in pursuit of optimized bottom lines for its shareholders, industry has commoditized and devalued skilled employees while simultaneously abdicating its social contract for the education of its most precious resource, its future workforce.

Now both parties decry the inability of the government to adequately fund the education system each abandoned in their rush to self aggrandizement.

What can post-secondary education do? (Get real …)

- Invite dedicated informed industry stakeholders onto curriculum advisory committees. Listen to them, but listen harder.
- Throw away your laminated lesson plans. Just because it was the right thing to teach yesterday, does not mean it is relevant today.
- Get involved with industry standards committees.
- Get out of the tower and get involved in current industry trends. Ask the following questions: What is industry doing? What do they need from us? How do we deliver?
- Quit making every student so specialized, limiting their value to employers.
- Get more practical and less theoretical. There must be a balance between the two.
- Start teaching technicians across the technical spectrum (e.g., mechatronics).
- Take over the technical training responsibilities abandoned by the secondary education system.

What can industry do? Invest in the future workforce that will make you successful.

- Stop outsourcing.
- Get involved in curriculum advisory committees at your local technical/community college or university.
- Open your facility to educators for industry co-ops during the summer.
- Badger your legislators to support education funding for skilled technician training.
- Quit turning a blind-eye to the closure of hands-on education programs in secondary schools.
- Start valuing technicians and technologists as much as you value engineers. (We hope.)

Effective technical education and the development of technical people by companies are serious problems that demand serious action. Forming a circular firing squad will not get the job done. This is not a gradual decline into mediocrity we face, but an ever steepening spiral into economic malaise.

We are rapidly approaching the tipping point. The day will come when some national security threat will wake us from our service economy hangover only to find the educational infrastructure needed to support a nimble, technologically-advanced response has fallen into such a state of neglect that it will collapse under the demand of the hour.

As Warren Buffett said: "You never know who's swimming naked until the tide goes out." Regarding the skill shortage, not only are we naked, but the global bully on the beach is threatening to kick sand in our works.

ABOUT THE AUTHOR

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Israel to build civilian nuclear plants

Israel said in March that it intends to develop civilian nuclear plants for energy, offering to build one as a joint project with Jordan, under French supervision.

According to *The New York Times*, the Israeli infrastructure minister, Uzi Landau, said Israel wanted a cleaner, more reliable source of energy than the large amounts of coal now imported. He said regional cooperation on civilian nuclear power could help bind the Middle East.

Jordan, however, said any such cooperation was premature before a settlement of the Israeli-Palestinian conflict.

Iran, already subject to sanctions by the United Nations Security Council, insists that its nuclear program is purely for civilian purposes, but Western governments believe its intentions are military. Still, Israel's announcement here may further complicate efforts to get the Security Council to impose new sanctions on Iran.

Israel has never admitted that it has nuclear weapons, and it has refused to sign the Non-Proliferation Treaty. Israel is a member of the International Atomic Energy Agency, and Landau said any nuclear power plant would be subject to international safeguards.



Israel has chosen a location in the northern Negev desert. "In a region like the Middle East, we can only depend on ourselves," Landau said. "Building a nuclear reactor to produce electricity will allow Israel to develop energy independence."

China, India give go-ahead to climate deal

China and India have given their qualified approval to the Copenhagen climate accord calling for voluntary limits on greenhouse gas emissions, according to *The Associated Press*.

More than 100 countries had earlier responded to a request to be "associated" with the nonbinding agreement brokered by President Barack Obama at the climate change summit in December.

But the delay in replying by the world's two fastest-growing polluters had raised concern the accord could be rendered meaningless, even though India and China were among a small group of nations that negotiated the deal.

China's one-sentence note to the U.N. climate change secretariat in mid-March said it agreed to be listed in the accord, which was seen as weaker language than asking to be associated with it.

Scandal prompts China to boost food safety

China will step up food safety efforts in the wake of a massive dairy scandal, expanding supervision to reach more of the country's countless small farms, an agriculture official said.

Vice Minister of Agriculture Wei Chao'an said agricultural officials at all levels are working this year "to prevent any large-scale food safety crises," according to *Manufacturing.net*.

Wei said China was working to bring more farms under better supervision, a challenge in a vast country where some rural areas are still very poor.

"Our agricultural products overall are safe and of high quality, but we must also recognize that while we transition from traditional to modern farming, many of our operations remain scattered, production methods are still backward, and our supervision lags behind," Wei said.

A ministry statement said the government promises to "implement quality and safety monitoring programs targeting raw and fresh milk, and strengthen supervision of purchase stations for raw and fresh milk."

Despite tightened regulations and increased inspections on producers, melamine-tainted milk products have recently shown up repackaged in several places around the country. Melamine, which can cause kidney stones and kidney failure, and is used to make plastics and fertilizers, has also been found added to pet food and animal feed.

Army launches apps contest

The U.S. Army launched its "Apps for the Army" (A4A) contest in March, open to active-duty, Reserve and National Guard personnel, and civilian employees. The service seeks good web and mobile software applications that can be used throughout the Army.

Lt. Gen. Jeffrey Sorenson, the Army chief information officer, said the purpose of the contest is "to encourage smarter, better, and faster technical solutions to meet operational needs."

The Army will distribute a total of $30,000 in prizes to winners of the contest, which is probably much less than it would cost to pay a contractor to write a thousand lines of code or an entire application. The top app submissions will be recognized at the LandWarNet Conference in August.

The Army said A4A apps that will be considered are ones that tackle "distributed training, battle command, career management, continuing education, or news and information distribution." A4A apps must be submitted by 15 May.

Key design components of final control elements

A final control element is the device manipulated by a control loop to affect the process, principally by means of changing a flow. Final control elements are an essential part of nearly every process control system. Without final control elements, there is no way of controlling the process. We could not change operating points or correct for disturbances. There may be several layers of control loops, but it is usually a flow that a final control element ends up changing in a process. The most notable exceptions are heater or electrode current and mixer speed.

THE BRASS TACKS

- The deadband, resolution, speed, and turndown of final control elements determine control system performance.
- Whether a valve or variable frequency drive has a better dynamic response depends on the application and adherence to best practices.
- Special variable frequency drive cables and installation considerations are needed to prevent damage and interference from electrical noise.

By far, the most common final control element is the control valve, with its attendant positioner, actuator, and other components. Variable speed peristaltic pumps are used for the exceptionally small flows of bench top and pilot plant operations. Variable speed positive displacement pumps are used for small additive and reagent flows in production. For large flows in plants and powerhouses, variable frequency drives and dampers are sometimes used instead of control valves to reduce capital and operating costs.

Axial and centrifugal blowers, fans, and pumps are used for the flow ranges normally associated with gas and liquid streams in industrial plants. A variable frequency drive (VFD), particularly in large utility flow applications, can save energy by the elimination of a control valve and its pressure drop. However, the energy savings is usually overestimated for process streams by not taking into account the service time and efficiency at low flow and the loss in turndown due to static head.

A damper can reduce the cost of the final element or fit in a non-circular duct. Dampers are commonly used in HVAC systems, boilers, furnaces, and scrubbers to manipulate air and vent gas flows. Dampers have a lower pressure drop than a control valve, but generally the performance (e.g., rangeability, resolution, sensitivity, speed, and seal) of a damper is not as good as a control valve. The leakage and limited dynamic response and materials/ruggedness of construction of dampers relegate their application to mostly utility and vent systems.

Valve design, dynamics

The shaft of the actuator and the stem of the internal closure component (plug, ball, or disk) of the control valve are normally separate. The closure component may be cast and forged with the stem or the stem may be connected during valve assembly. The actuator shaft moves the stem that moves the closure component. (While "shaft" and "stem" are more appropriate terms for the actuator and the closure component, respectively, in practice the terms "stem" and "shaft" are used interchangeably.) The amount of play (looseness or gap) in the connections between the shaft, stem, and closure component is backlash that creates deadband and determines, in part, how well the valve will respond to small changes in signal. Excessive seal friction of a closure component that is rotated (e.g., ball or disk) can result in shaft windup. The location and type of connection of the positioner feedback mechanism for valve travel determines whether the positioner is seeing the response of just the actuator or the actual response of the closure component.

Previous methods of testing valve response involved making much larger changes in the valve signal than would normally be made in closed loop control. Most valves will look OK with these large changes in requested position. In service, the change in controller output from scan to scan is generally small (e.g., < 0.2%), except during the start of an operation or process. For small changes in valve signals, the resolution limit from sticktion and deadband from backlash that prevent a good response and create a sustained oscillation (limit cycle) are observable. Current test methods established by the ISA-75.25.01-2000 (R2006) standard address the effect of step size on response.

Control valves with excessive sticktion, backlash, and shaft windup can actually increase process variability when the loop is in automatic by the creation of oscillations from the continuous hunting of integral action to find a position it cannot attain exactly.

Smart digital positioners with a good closure component measurement have the sensitivity and tuning options to mitigate the consequences of stick-slip and backlash by fast feedback control. Built-in diagnostics can pinpoint problems such as packing friction besides monitoring the dynamic response of the valve.

Sliding stem (globe) valves have the least amount

of deadband because of the direct connection between the actuator shaft and trim stem, and low trim friction. For rotary valves, connections can be problematic since there is the need to convert the linear motion of a piston or diaphragm shaft to rotary motion and the changes in the effective lever arm length. Rotary valves originally designed by piping valve manufacturers for on-off or manual operation often have a non-representative position measurement and a degree of excessive backlash and shaft windup that cannot be corrected by a positioner.

Valve best practices

For best performance, users should consider the following during the specification of control valves:

- Actuator, valve, and positioner package from a control valve manufacturer
- Digital positioner tuned for valve package and application
- Diaphragm actuators where application permits (large valves and high pressure drops may require piston actuators)
- Sliding stem (globe) valves where size and fluid permit (large flows and slurries may require rotary valves)
- Low stem packing friction
- Low sealing and seating friction of the closure components
- Booster(s) on positioner output(s) for large valves on fast loops (e.g., compressor anti-surge control)
- Online diagnostics and step response tests for small changes in signal
- Dynamic reset limiting using digital positioner feedback

VFD cable problems

Belden Inc. has studied the radiated noise from cables between the VFD and the motor. Unshielded VFD cables can radiate 80V noise to unshielded communication cables and 10V noise to shielded instrument cables. The radiated noise from foil tape shielded VFD cables is also excessive. A foil braided shield and armored cable performs much better. Still, a spacing of at least one foot is recommended between shielded VFD and shielded instrumentation cables. The cables should never cross. As a best practice, separate trays to isolate VFD and instrumentation cables should be used to avoid mistakes during plant expansions and instrumentation system upgrades.

VFD turndown

Since the inverter waveform is not purely sinusoidal, it is important to select motors that are designed for inverter use. These "inverter duty" motors have windings with a higher temperature rating (Class F). Another option that facilitates operation at lower speeds to achieve the maximum rangeability offered by a pulse width modulation (PWM) drive is a higher service factor (e.g., 1.15).

The turndown of a VFD could drop to 4:1 for the following systems:

- Older VFD technologies such as 6-step voltage (excessive slip at low speed)
- Systems with a high static head (flow plummets to zero at a low speed)
- Operation on the flat portion of the prime mover curve (cycling at low speed)
- Hot gases (motor overheats at a low speed)

VFD controls

The turndown (rangeability) of a VFD can be increased by ensuring the pump head is large compared to the static head, by using PWM inverters, and by dealing with the heating problems at low speeds. Turndown also depends upon the control strategy in the variable frequency drive.

Which is faster: A valve or VFD?

Exceptionally fast loops can ramp off-scale in milliseconds. These loops have essentially a zero process dead time and may have a high process gain due to a narrow control range (e.g., fractional inches of water column for furnace pressure). These loops require DCS scan times of 0.05 to 0.1 seconds. Special fast scan rate digital controllers or analog controllers are needed. DCS scan time requirements of 0.2 seconds or less signify a VFD opportunity. A properly designed VFD has no measureable dead time, while control valves and dampers take anywhere from 0.2 to 2.0 seconds to start to move. For example, an incinerator pressure and polymer pressure loop that could get into trouble in less than 0.1 second required a VFD and analog controller to stay within the desired control band.

A VFD has a negligible response time delay unless a deadband or dead zone is introduced into the drive electronics to slow response to process measurement noise, or if a low resolution input card is used. A control valve or damper has a dead time proportional to the resolution limit (e.g., from stiction and windup) and dead band (from backlash and windup) divided by the rate of change of the process controller output. For large or fast changes in signal, this dead time disappears.

VFD best practices

With a VFD, a tachometer or inferential speed feedback signal should be sent to the process controller in the DCS that is sending the signal to the drive. The speed feedback should be used in a similar way to the position feedback from a digital positioner to prevent the process controller output from changing faster than the VFD can respond. The use of the dynamic reset limit option for the loops in the DCS can automatically prevent the process controller from outrunning the response of any type of final element. For best performance, users should consider the following during the specification and implementation of variable frequency drive systems:

- High resolution input cards
- Pump head well above static head on-off valves for isolation
- Design B TEFC motors with class F insulation and 1.15 service factor
- Larger motor frame size
- XPLE (cross-linked polyethylene) jacketed foil/braided or armored shielded cables
- Separate trays for instrumentation and VFD cables
- Inverter chokes and isolation transformers
- Ceramic bearing insulation
- Pulse width modulated inverters
- Properly set deadband and velocity limiting in the drive electronics
- Drive control strategy to meet rangeability/speed regulation requirements
- Dynamic reset limiting using inferential speed or tachometer feedback

Source: *Essentials of Modern Measurements and Final Elements in the Process Industry: A Guide to Design, Configuration, Installation, and Maintenance* by Gregory K. McMillan (ISBN: 978-1-936007-23-3) www.isa.org/finalelements

A standard grows up: The evolution of ISA's standard on alarm management (ISA-18.2)

By Todd Stauffer

On 23 June 2009, ANSI/ISA-18.2, "Management of Alarm Systems for the Process Industries" (ISA-18.2), was released. As with many standards, completion of ISA-18.2 entailed significant effort from a cross-functional team of volunteers representing end users, suppliers, consultants, integrators, and the government. The ISA18 committee labored for more than five years, turning out eight drafts of the standard and reviewing/resolving almost 4,000 comments. Release of a standard, however, is just one stage in its life. Performance-based standards define the "what," but not the "how." Application guidelines and examples, the "how," are needed to support wide-spread adoption by industry.

Overview of ISA-18.2

ISA-18.2 provides a framework for the successful design, implementation, operation, and management of alarm systems. It contains guidance to help prevent and eliminate the most common alarm management problems, as well as a methodology for measuring and analyzing performance of an alarm system. The standard is organized around the alarm management lifecycle. The key activities of alarm management are executed in the different stages of the lifecycle. The products of each stage are the inputs for the activities of the next stage.

Defining an alarm

Several of the most important principles of alarm management are highlighted in the definition provided by ISA-18.2.

An alarm is …

- **An audible and/or visible means of indicating—**There must be an indication of the alarm. An alarm limit can be configured to generate control actions or log data without it being an alarm.
- **To the operator—**The indication must be targeted to the operator to be an alarm, not to provide information to an engineer, maintenance technician, or manager.
- **An equipment malfunction, process deviation, or abnormal condition—**The alarm must indicate a problem, not a normal process condition.
- **Requiring a response—**There must be a defined operator response to correct the condition. If the operator does not need to respond, then there should not be an alarm.

Benefits of ISA-18.2

A well-functioning alarm system can help a process run closer to its ideal operating point, prevent unplanned downtime, and keep the process running safely. Poor alarm management can affect an operators' performance by making it more difficult for them to detect, diagnose, and respond to each alarm correctly and within the appropriate timeframe. Following the alarm management lifecycle of ISA-18.2 can go a long way toward eliminating and preventing common alarm management problems such as:

- Nuisance alarms
- Chattering & fleeting alarms
- Stale alarms
- Alarms with no response
- Alarms with the wrong priority
- Redundant alarms
- Alarm floods

Next steps

As part of the continuing evolution of ISA-18.2, a series of ISA18 technical reports (TRs) is being developed to help alarm management practitioners put the requirements and recommendations of ISA-18.2 into practice.

Alarm Philosophy (Technical Report 1): The cornerstone of an effective alarm management program is the alarm philosophy document, which defines how a company or site will execute alarm management. TR1 will define roles and responsibilities, how to classify and prioritize alarms, what colors will be used to indicate an alarm in the HMI, and management of change procedures. It will also establish key performance benchmarks (e.g., acceptable alarm load for the operator).

Alarm Identification & Rationalization (TR2): This TR will describe how to evaluate whether something should be an alarm, and how to set its priority, classification, and limit by considering time to respond, process dynamics, and potential consequences.

Basic Alarm Design (TR3): This TR will provide guidance and application examples covering the selection and configuration of alarm attributes (types, deadbands, and delay time).

Enhanced and Advanced Alarm Design (TR4): This TR will describe how to deliver information to the operator to help formulate a response, to modify alarm attributes dynamically based on operating state, to address events that trigger multiple alarms, to use model-based predictive alarming, and to redirect alarms outside of the control room.

Alarm Monitoring, Assessment, and Audit (TR5): This TR will provide guidance on how to measure, analyze, and improve alarm system performance through evaluation of key performance indicators.

Alarm Systems for Batch and Discrete Processes (TR6): This TR will specifically address how the standard applies to batch and discrete processes. It will provide guidance on how to deal with the nuances of managing alarms associated with batch and discrete processes.

Looking for good men, women

If you are interested in contributing your knowledge and experience to the TR development effort—and in gaining from the knowledge and experience of your professional colleagues at the same time—please contact ISA18 co-chairs Nicholas Sands (Nicholas.P.Sands@USA.dupont.com) or Donald Dunn (Donald.Dunn@aramcoservices.com).

ABOUT THE AUTHOR

Todd Stauffer (tstauffer@exida.com) is an alarm management consultant for exida and a voting member of the ISA18 committee. He is co-chairing the development of TR3 on basic alarm design.

Engineers—re-engineer yourself

By Jim Pinto

Decades ago, technology brought the era of "specialization"—knowing more and more about less and less. To advance faster, you had to focus. But in today's global environment, new developments have accelerated to where companies must generate winning strategies beyond narrow technical advantages. Broad leadership vision and teamwork have become important.

Engineering has an image problem. Surveys show the public is not aware of what engineers do, beyond being involved in construction of machines and buildings. Most people tend to think of engineering as being a job concerned with objects and gadgets rather than people. Actually, those ideas start with engineers themselves. It is their self-image.

Narrow focus = tunnel vision

Engineers tend to focus on engineering, rather than the overall, broad picture. And this limits their leadership potential. Most engineers do not want to be managers because they recognize leadership involves many things beyond the technical details they enjoy. They feel they should stick with what they know rather than branch off into the grey goop of people interface. Or even worse, marketing or sales, which engineers jokingly call "the dark side."

Did you know very few company chief executives are engineers? Even in technology companies, the top gun is typically a marketing person, followed (in order of probability) by finance, then sales, then operations (manufacturing), and last engineering. Especially in engineering companies, this is strange because, in my opinion, it is easier to teach an engineer about marketing than it is for a non-technical sales or marketing person to learn engineering. Engineers who advance to executive leadership can make a big difference.

I am an engineer, and so I feel I can discuss these things for and about engineers. Early in my engineering career, I was as frustrated at the lack of leadership around me. Most people seemed happy to be part of success, but did not take responsibility when things went wrong. Then I realized, directly or indirectly, I was part of the problem. Instead of kicking back to blame others, I started to find ways to become part of the solution. I started taking responsibility and got promoted. I discovered this truism, "I looked for a leader, and found myself!"

Success demands many disciplines

Engineering is a detail-orientated job. The design of products and systems entails a host of details that must be integrated. And so, engineers are usually narrowly focused, trusting in the old adage, "Build a better mousetrap, and the world will beat a path to your door."

The truth is the better mousetrap does not sell itself. Before the design is even contemplated, you must know the target customer. The "to do" list for design optimization must include the important market requirements. This involves comparing available products, reviewing competitive features, advantages, and benefits, finding out whether engineering can offer something superior, reviewing sales channels, and coverage of key geographical market areas.

Good engineering must be involved with all of these things to understand how and why the design specifications have been generated before the real engineering can commence. If you have a good understanding of the marketing requirements, plus the follow-on manufacturing, quality, sales, and distribution needs, then you are a good engineer. This is what I call "total concept engineering."

Re-engineer yourself

If you are an engineer and want to move ahead in your management career, you need to be constantly re-educating yourself in other disciplines. Here are some positive ideas on what you can do to re-engineer yourself.

- Make sure you re-invent yourself on a daily basis. Start digging into things that affect your job and your company, beyond just engineering. If you are proud of the products you helped develop, find out what it takes to make those products successful.
- Read the corporate business plan. Make an effort to understand other departments' goals and objectives. Dig into the things that help to make your company successful. Most good companies will welcome your broader involvement. If they do not, go up the chain till you get to the leader who will encourage you to understand more.
- Do not get stuck on narrow details. Go beyond your own projects, and see how everything contributes to the company's goals. Success involves identifying the results required and knowing the right steps, which includes recognizing the wrong steps. Ask questions to gain a clear understanding of what it takes to accomplish the overall objectives effectively.
- Become more proactive by finding productivity improvements and selling management to implement those changes. Take time to talk with marketing on product requirements and specifications; work with manufacturing to optimize production methods and costs; come up with ways to minimize hardware inventory by developing selection options; be pro-active in the specifications, to beef up the advantages. There are dozens of ways to dazzle the customers, so keep looking for them.
- Get to know your customers. These are the people (inside or outside your company) for whom you are doing the work. Go with sales people to visit customers to find out what they are buying and why. Satisfaction will bring customers back to generate success for your company.

Re-engineer yourself. You will enjoy the growth and success that this will bring.

ABOUT THE AUTHOR

Jim Pinto is an industry analyst and founder of Action Instruments. You can e-mail him at jim@jimpinto.com or view his writings at www.JimPinto.com. Read the Table of Contents of his book, *Pinto's Points*, at www.jimpinto.com/writings/points.html.

Thriving during economic downturn by building real-time enterprise

By Peter G. Martin

Part 2 of a two-part series

The previous article in this series (www.isa.org/intech/workdev_201002) discussed the role of people, information, and technology to enhance the performance of existing plant assets in today's challenging economic environment, and how companies can cope by employing real-time techniques in enterprise management using resources they may already have. This article details the real-time business approach to measurement, employee empowerment, and operations, ultimately leading to real-time profit optimization.

Although the concept of a real-time enterprise might seem daunting, the basics of controlling and improving such an enterprise are much simpler than they may initially appear.

A real-time enterprise requires business and operations information to be available to operations personnel and management in real time, but traditional IT systems are not designed to provide information so frequently. They are optimized around monthly, weekly, or at best, daily schedules, and they typically do not contain data that reflect the real-time operation of the business. It is impossible to take monthly data and extract minute-by-minute guidance from it.

On the other hand, automation systems were designed from inception to operate in real time. They are also connected to a real-time process instrumentation that reflects everything happening in the plant second by second and can be thought of as the real-time database of the industrial operation.

Granted, this "database" is difficult to use from a business perspective, containing information such as flows, levels, temperatures, pressures, speeds, and chemical compositions, but there is a clear relationship between these basic process variables and the required real-time business variables. Using real-time business information, such as current energy costs that can be retrieved directly from utilities in electronic format, basic process data can be transformed into required business and operating data in real time, allowing plant personnel to make real-time decisions that improve the plant performance. And this typically requires no additional capital investments.

All it requires is talent that can use existing plant automation and information systems in a new and different way to apply real-time operational and business information. When the real-time business data of an industrial operation is utilized to drive business performance improvements in this manner, the resulting operation is referred to as a real-time enterprise.

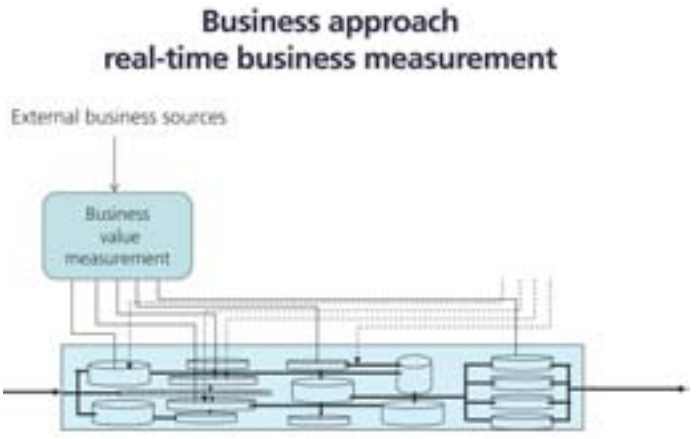
Although the concept of a real-time enterprise might seem daunting, the basics of controlling and improving such an enterprise are much simpler than they may initially appear. The key is to build on the knowledge engineers have developed over the past 50 years in controlling real-time production processes.

As with any control challenge, the first component that needs to be developed is the measurement of the variables to be controlled in the time frame necessary for control. In the case of today's business variables of industrial operations, the timeframe has to be in real time.

The challenge is how to measure business variables in real time. The business variables of an operation are commonly managed through the company's ERP system, which is anything but real time. Therefore a new database is required that will underpin the creation of the real-time business variables.

Fortunately, in modern industrial plants that are under automatic process control the database exists, even if it is not initially obvious. The real-time data sourced by the hundreds of process sensors installed in most plants provides an ideal database for the development of the real-time business metrics. If the equations of the necessary operational measures (key performance indicators) and financial measures can be determined, typically an experienced engineer can develop models of those measures primarily using the plant sensors as source data. Often additional external information may be required, such as the current price of energy. These models can execute right in the controllers of the plant automation system, providing the necessary real-time business measures.

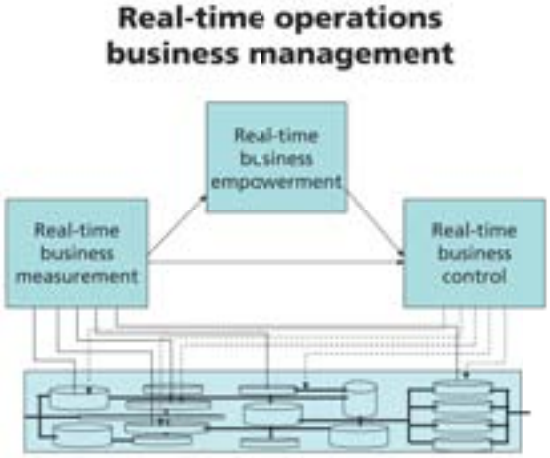
Once the real-time business measurements have been developed, the second step is to move to bring these measurements under feedback control. As with the early process control systems, the best and easiest starting point is the move to manual feedback control by using the plant operators to take control action. In early process control systems, this was accomplished by assigning an operator to control a specific process variable by turning a hand valve and empowering the operator to make the right decisions through a gauge that indicated the cur-



rent value of the process variable.

For business variables, a similar approach can be taken. Plant operators can be empowered though the creation business gauges in the form of dashboard displays of the real-time business variables.

These dashboards have to be carefully developed and contextualized to the skills and experience of each individual employee involved in the control of the business. This careful process results in a set of manual feedback business control loops focused on production value, energy cost, material cost, environmental integrity, and safety.



Some business variables are beginning to fluctuate so rapidly that it is becoming very difficult for operators to control them through a manual feedback control system. In these instances, automatic feedback controllers of the business variables will need to be developed. The algorithms for these business controllers may not be as straightforward as the general purpose PID algorithm used in process control.

But careful analysis of each business control problem can often result in the effective development of a special business control algorithm. The result is an automatic business control loop.

Humans also tend to have difficulty with business variables that respond too slowly to changes they make. For example, in a refinery an operator may change a set point to drive a business result; but due to the dead time in the process, the actual result may not be realized for hours. By the time it is, the operator may be at home eating dinner and may never find out the impact of the change. In these cases, technologies such as online simulators may be deployed. When the operator makes a change, the simulator can go into fast-forward mode and immediately let the operator know what the impact of the change will be when it worked through the process.

Once each of the five key business variables of an industrial operation—production value, energy cost, material cost, environmental integrity, and safety—are brought under control, the final step is to optimize the profitability of the plant in real time. The simplified plant profit real-time optimization model shows three business objectives—maximizing production value, minimizing energy cost, and





minimizing material costs—constrained by two key business constraint functions, environment and safety.

This model clearly shows optimizing plant profitability is a multiple objective optimization problem. Focusing on any single objective in deference to the others would result in a sub-optimized business. Unfortunately, effective mathematical approaches to resolve multiple objective, real-time optimization problems do not yet exist. Studies have found a human with a moderate education and the right information in the right timeframe will earn how to solve a multiple objective optimization problem through experiential learning—so manual real-time profit optimization is available today.

Every aspect involved in moving to real-time profit optimization involves utilizing the installed plant capital investments in the form of automation and information technologies more effectively to solve new kinds of business problems. In many cases, this can be accomplished without the need to acquire any new technology.

Most control and process engineers have the background and experience to make great strides toward business performance improvement by using the models presented, along with their traditional knowledge and skills as control engineers, and the installed automation and information technologies.

This is a difficult economic time for industry. But such times offer new opportunities for improvements such as described here. With the reduction of capital projects, industrial companies can mobilize their existing human assets to build real-time operations business improvement programs using their automation and information assets. As a result, not only will they ride the downturn with less disruption, they will also be in a great position to capitalize when the economic upswing comes. The time is now. The opportunity is ripe.

ABOUT THE AUTHOR

Peter G. Martin Ph.D. is vice president for business value generation at Invensys Operations Management. Martin has spent three decades in the automation industry, culminating with the development of commercially-applied dynamic performance measurement technologies and methodologies. An established author and industry speaker, he received the ISA Life Achievement Award in 2009 for his work in performance measurement.

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PAKS committee gears up for Control Systems Engineer licensing

Who decides what a licensed Control Systems Engineer (CSE) needs to know? Educators, legislators, industry? Envelope please … it is YOU, the practitioners, based on your employers' and clients' needs.



The PAKS committee is under the Professional Development Department of ISA.

How are the knowledge areas determined? Every six to eight years a Professional Activities and Knowledge Study (PAKS) survey is held by a sponsoring society. For a CSE, this is the International Society of Automation (ISA), and it is done in conjunction with the National Council of Examiners for Engineering and Surveying (NCEES). The last PAKS survey for the CSE was performed in 2001, and now it is time to do it again.

To perform this survey, the PAKS committee meets to review the control system specification framework, develop the survey questions, review the survey prior to release, review the results of the survey, and then review any revisions to the CSE Exam Specification. The first exam under the new specification, which is called the *Anchor Exam,* is then assembled and tested. After the Anchor Exam is given, a Cut Score Panel is convened. The Cut Score Panel actually takes the exam, discusses the questions, and recommends a passing score to NCEES. Once these tasks are completed, all future CSE exams will be referenced back to the Anchor Exam until the next PAKS survey is undertaken.

The purpose of licensing engineers is to protect the public. Each state and territory controls licensing via the enactment of laws and rules. One cannot call themselves an "Engineer" or offer "engineering services" unless they are licensed in that state where they are practicing. The PAKS survey is the method used to help determine what knowledge and activities licensed engineers must know to be minimally competent. The most complete determination of this knowledge and activities is when all practicing facets of our profession are involved and the committee overseeing the PAKS represents that diversity.

The 2010 PAKS committee is made up of 19 members from nine different states, six of which are in the top 10 states for CSE licensing. The members are experienced in the following industries:

- Aerospace
- Chemical
- Education
- Food & Beverage
- LNG
- Oil & Gas (Upstream)
- Petrochemical
- Pharmaceutical
- Polymers
- Power
- Pulp & Paper
- Refining
- Safety Systems
- Steel
- Textiles
- Water & Wastewater

When the first CSE Specification was developed in the early 1990s, it was based on the role of the instrument engineer of that period. From one viewpoint, it was a world based on a renaissance engineer specification—one who could do it all. The framework was divided into the following:

- Measurement
- Signals and transmission
- Final control elements
- Control systems analysis
- Control systems implementation
- Codes, standards, and regulations

Today, a CSE's world has evolved into one with many more specialists and fewer renaissance engineers. The exam framework also needs to evolve to meet current practice. One thought is to use the framework definition of a control system found in ISA84, Application of Safety Instrumented Systems of the Process Industry (a.k.a. IEC-61511 Functional Safety of Safety Instrumented Systems for the Process Industry Sector), which defines a control system as the input devices, final control elements, basic process control system, and the safety instrumented system. From this definition, the proposed framework may be:

- Measurement (Inputs)
- Final control elements
- Process control system
- Safety instrumented systems
- Standards and codes

To determine which item is important to which practitioner, one of the first questions to be asked is, "what is your major area of practice?" The suggested choices include:

- Instrument engineer, who specializes in measurement and final control areas
- Process control engineer, who specializes in distributed control systems, remote-terminal units, programmable logic controllers, human-machine interfaces, advanced process controls, and other process control applications
- Safety system engineer, who specializes in safety instrumented system applications

By analyzing the survey information in these major areas of practice with questions that examine the aspects of our work, the results will be used to update the CSEs' exam specification.

For this year's PAKS to best represent what CSEs do and how they serve the public, we strongly encourage you to participate in the survey. As this survey develops, additional information on when, where, and how you can participate will be published. For more information on the PAKS survey, contact Dalton Wilson (dwilson@isa.org).

RESOURCES

CSE licensing information
www.isa.org/link/CSE

PE exam information
www.ncees.org/Exams/PE_exam.php

Documenting skills is value-add

ISA certification provides an objective, third-party assessment, and confirmation of a person's skills. It gives manufacturing and factory staff the opportunity to differentiate themselves from their peers and gain recognition. *InTech* covers two certification areas in this monthly Certification department

ISA Certified Automation Professional (CAP) program

Certified Automation Professionals (CAPs) are responsible for the direction, design, and deployment of systems and equipment for manufacturing and control systems.

CAP question

Nonincendive-rated field wiring is allowed to be used in which of the following combinations of hazardous area classifications?

A. Class 1, Division 2 and Class 1, Zone 0
B. Class 1, Division 1; Class 1, Division 2; Class 1, Zone 1 and Class 1, Zone 2
C. Class 1, Division 2; Class 1, Zone 1 and Class 1, Zone 2
D. Class 1, Division 2; Class 1, Zone 2 and Non-hazardous

CAP answer

The correct answer is **D**. Nonincendive is allowed in Class 1, Division 2; Class 1, Zone 2, and Non-hazardous rated areas. Use in higher hazard areas, such as Division 1, Zone 1, or Zone 0 areas, is not allowed.

Reference: NFPA 70, National Electrical Code 2005 or 2008.

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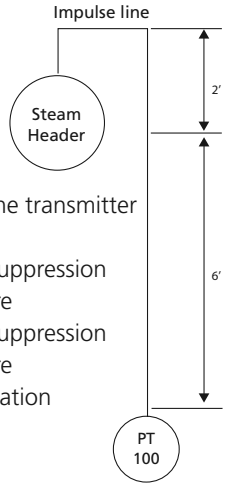
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ISA Certified Control Systems Technician (CCST) program

Certified Control System Technicians (CCSTs) calibrate, document, troubleshoot, and repair/replace instrumentation for systems that measure and control level, temperature, pressure, flow, and other process variables.

CCST question

A gauge pressure transmitter that measures the pressure in a 150 # high pressure steam header is mounted 6 feet below the center line of the header. The tap for the impulse line connects to the top of the header and rises 2 feet above the header center line, extends horizontally for 3 feet, and then drops down to the transmitter. In order to read the pressure in the steam header correctly, the transmitter output must be:



A. Calibrated for suppressed zero, the suppression equal to 8 feet of liquid head pressure
B. Calibrated for suppressed zero, the suppression equal to 6 feet of liquid head pressure
C. Calibrated for elevated zero, the elevation equal to 8 feet of liquid pressure
D. Calibrated for true zero

CCST answer

The correct answer is **A**, calibrated for suppressed zero with the suppression equal to the 8 feet of liquid height that is in the impulse line.

At zero gauge pressure in the steam line, you are essentially suppressing (pushing) the transmitter output back down to zero after the system reaches equilibrium with the 8 feet of impulse line full of liquid. The transmitter will read that impulse line liquid head pressure plus any pressure exerted on top of the liquid. 6 feet of liquid head suppression is incorrect as the impulse arrangement will cause 8 feet of liquid head to accumulate. Elevated zero is incorrect as that adjustment is used to adjust for negative pressure offsets resulting from the transmitter being mounted above the zero reference point (high pressure tap point) and where a liquid in the impulse line or a capillary system would exert a negative pressure.

Reference: Thomas A. Hughes; *Measurement and Control Basics, 4th Edition*, ISA Press, 2007.





Focus on valves and actuators

Modular linear actuators



The Stainless Steel Modular Linear Actuator Series (QLZE / QSZE / QST-KE) Positioning System is engineered with a stainless steel cover band that wraps over the square aluminum profiles, which creates an optimal solution to combat caustic wash-down issues (acid solvent/acid bath cleaning). The stainless steel housing sheet (thickness 0.37mm, material 1.4301) protects the aluminum profile from exposure to damage due to clogging and pitting, thus protecting the deep structures of the extrusion and extending the system's life. The stainless steel design requires no bellows to cover the extrusion which typically wears out over time.

Customized to customer specification from the company's standard line and available in ball screw and belt drive options, the linear actuator series is available in a number of sizes (60 / 80 / 100) and provides an exceptional performance solution for precision and commercial caustic wash-down applications.

The standard single-carriage linear actuators offer push-pull type directional loading that can handle dynamic maximum capacity carry loads up to 1,600 lbs of force and maximum thrust loads up to 1,100 lbs of force.

Fully assembled and configurable with Nook motors (ac/dc/servo/stepper) and stainless steel hardware including pulleys, limit switches, end mounts, etc., the standard fitting position of each in the series offers maximum stroke-lengths of 3,000 mm (without joints).

Nook Industries, Inc.
www.nookindustries.com

Cartridge valves



A line of manifold mount Cartridge Valves is an alternative to solenoid operators and stud mount valves. The Cartridge Valve is part of a family of direct-acting solenoid valves for air and liquid (including light oil) applications. These small yet powerful valves meet today's commercial, industrial, mobile, instrumentation and medical market needs. Economy models are available for less demanding applications.

The manifold mount Cartridge Valve is offered in two- and three-way models with a stainless steel body. It offers a space-saving approach, without manifold orifices to machine or press. Fully assembled, the valve offers no loose parts to assemble together—the sleeve, plunger, spring, and orifice are pressed together as one unit. Cartridge Valves are 100% tested for quality and durability. Available with all 204 and 304 coils, the valves are offered normally closed or open with a sleeve port size of 1/8". The valves' orifice sizes range from 3/64" through 5/32".

Parker Fluid Control Division
www.parker.com/fcd

Compact gauge valve



The compact gauge valve is designed with a smaller footprint and for maximum performance. The design provides quick, convenient access for the isolation and venting of pressure gauges. The valve can be used with the company's pressure gauges. The pressure gauges are positionable with tube adapter ends, eliminating threaded connections and leak points.

The smaller, lightweight footprint reduces the need for supports, which place additional stress on a system. A streamlined body features the company's tube fitting end connections for leak-tight performance, reduced installation time and cost, plus robust tube grip and vibration resistance.

The design incorporates a purge valve for easy bleeding of trapped fluid pressure between the valve seat and gauge upon shutoff. The purge valve is machined directly onto the body, eliminating potential leak points while allowing the user to safely release the fluid before removing the gauge. A permanently assembled purge cap is crimped to the valve body for operator safety and to prevent accidental disassembly.

Available with either ½ in. or 12 mm tube fitting end connections, the compact gauge valve is constructed from 316 stainless steel. The valve is rated for temperatures up to 450°F (232°C) depending on stem and packing.

Swagelok Company
www.swagelok.com

Butterfly valve



An expansion of the large diameter size range for the Type 567 Butterfly Valve now includes sizes 14-16 inches. The new size range features a unique double eccentric off-center design and excellent chemical compatibility. The 567 Valve's unique double eccentric design features an off-center shaft that allows the disc to completely disengage from the disc seal, even when partially open. This results in reduced seal friction for longer service life and minimal maintenance compared to non-eccentric valve styles. The double eccentric design requires only about half the torque of a traditional boot design, which decreases wear on the disc seal to further enhance product life.

GF Piping Systems
www.gfpiping.com

Linear voice coil housed actuator



The Linear Voice Coil Housed Actuator features its own shaft and bearings as well as an integrated AMP connector, which simplify OEM installation. The presence of the integrated AMP connector eliminates flying leads and the need for terminals. The only part needed by the customer for actuator installation is a mating female connector. The actuator, manufactured with a linear bearing system, provides an extremely large number of cycles (tests have proven performance up to 10 billion cycles). A flex circuit inside the device also contributes to high reliability of operation. Designed in a durable package measuring 2.75" (70 mm) in diameter and 5.2" (132.7 mm) in length, its large size allows for a Peak force capability of 60 lbs. (267 N).

BEI Kimco Magnetics
www.beikimco.com

Industrial radios

The 802.11n Industrial Hotspot solutions leverage the latest technology to provide greater flexibility and performance to a broad range of manufacturing and production applications. The 802.11n Industrial Hotspots improve performance for high-bandwidth video/voice applications and high packet-rate control applications. They provide better signal sensitivity and range in environments like mobile factory settings. Options are available to suit the application, including Single (RLXIB-IHN), Dual (RLXIB-IH2N), and Watertight Dual (RLXIB-IH2N-W) Industrial Hotspots. The radios utilize MIMO (Multiple Input, Multiple Output), a technology that uses up to three antennas to enable high-speed data rates up to 300 Mbps, providing advanced performance in industrial environments.

ProSoft Technology
www.prosoft-technology.com

Turbidity analyzers



The CE compliant TRCN440 series turbidity analyzers are available in three models. The TRCN441 is designed for medium to high turbidity ranges, and the TRCN442 targets lower turbidity ranges ideally from 10 to 100 NTU and is ideal for applications in the beverage industry, mining and water treatment. The TRCN443 is specifically designed for ultra low turbidity readings below 10 NTU for applications like drinking water and pure water. All models are compact in design, are light weight with the NEMA 4X (IP67), and include local indication, transmitter outputs, and control outputs.

OMEGA Engineering
www.omega.com

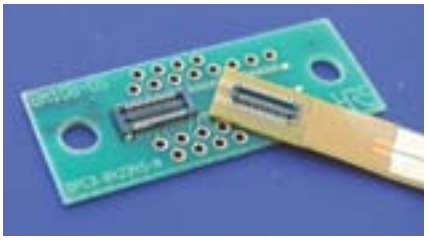
I/O server industrial PC



The I/O Server Industrial PC features an internal carrier card to interface a wide selection of related plug-in I/O modules. Designed specifically to work together, this combination of a rugged, fanless box computer and conduction-cooled I/O modules provides an integrated system for high-performance measurement and control projects. The first release in the I/O Server line, the Model IOS-7400, is equipped with an Intel Atom CPU and interface connections for peripherals and network devices. Users can insert up to four mezzanine IOS modules, in any mix, onto the slide-out carrier card to perform A/D, D/A, discrete monitoring/control, counter/timer, serial communication, and FPGA computing functions. The interface for up to 192 channels of field I/O is handled through four high-density connectors on the front panel for clean, easy cable access. Advanced thermal technology removes heat without open vents or fans for dependable operation from -30 to 75°C. Pricing for the I/O Server PC starts at $2195 while the twenty-plus IOS modules begin at $325 each.

Acromag
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Engineering automation? ... Just mash the button

By Ken Valentine

What does automation mean to today's modern engineering contractors? From the perspective of the automation engineering discipline, this means designing efficient systems with modern instrumentation, the most current process control systems with all the bells and whistles, IEC61511-compliant safety instrumented systems, cutting-edge analyzers and sample systems, and the most up-to-date communication system. Though the cost of a control system including equipment, engineering, and design is less than 3% of the total installed cost of most major projects, the opportunities for improved return on investment are greatest within the discipline in the form of advanced process control, energy savings, enterprise solutions, asset management, plant maintenance, medium and high fidelity process modeling, and operator training. This is how an optimized facility is achieved; but how do the contractor's engineering groups use automation to optimize their productivity?

Engineering automation in this context is defined as the software tools and processes used during the design phases of a project by all engineering disciplines, and it plays a very large factor in the design of a facility. Modern Engineering Contractors use some form of engineering automation in their design, from 3-dimensional (3-D) modeling to automatic generation of drawings using a data-centric environment.

When engineering contractors first started using personal computers tied together within networks, most automation was limited to creating small macros within a single electronically generated drawing. The computer replaced the drafting table, and the effort it took to generate a CAD drawing was about the same as producing a drawing with a T-square and lead holder. As engineers became more experienced with databases, they started to automate the generation of deliverables. As the computers and networks became more powerful, the automation software evolved, and data was put on servers able to handle larger amounts of data. Work teams became virtual, and projects were in operation worldwide.

The design process starts with the overall blueprint for the facility—the piping and instrumentation diagram, or P&ID. Today's "intelligent" P&ID has common data imbedded in the symbology that is initiated by the process engineer and used by all engineering disciplines to develop the facility. After the P&IDs are issued for design, the 3-D model and work in the engineering system design software begins.

Prior to computers, plastic models were developed to lay out facilities. This ensured all pieces of the facility would intersect accurately and the facility would fit onto the allowable real estate. However, even the best modelers could only achieve a design accuracy of a few inches. Today's facility constructability accuracy is dependent more on the quality of the field craftsmen, such as welders and pipefitters, than the dimensional accuracy of the model which is a fraction of an inch. Using today's 3-D model results in fewer engineering and field hours, fewer paper documents, and less waste for lower cost and "greener" design.

The 3-D model is a multi-discipline effort. The instrumentation and electrical disciplines model the lighting, raceway and conduit, junction boxes, electrical equipment, in-line instruments, local panels, remote buildings, and stand-mounted instruments. The documentation that goes to the fabricators and construction contractor are automatically generated by the software and delivered electronically. The 3-D model is then used by construction for planning, reporting, and progressing.

While the 3-D work is going on, data is being loaded into the instrumentation design platform, which will eventually produce datasheets, wiring interconnection reports, instrument indexes, cable schedules, and other deliverables. Data is imported into the control and shutdown systems for configuration. Much of the engineering documentation is viewable through the control system or plant-wide enterprise solution.

After construction is complete, owners use data generated during the engineering phase for their maintenance, asset management, and enterprise solutions.

So what is the ultimate goal that engineering contractors and clients want from engineering automation systems? The "Holy Grail" of engineering automation has always been to have an integrated software package that creates an "instant design"—otherwise known as "just mash the button," as a senior ISA member used to say. Even with the application of data-centric systems, is it feasible for engineers to develop a complete design that includes all the engineering deliverables that are needed? Owners would have to accept standardized designs based on common industry practices and specifications. In reality, few owners are willing to accept designs based on this type of standardized criteria. We are still a long way from our goal of achieving this level of engineering automation with the technology available today, but we are getting there with each step.

ABOUT THE AUTHOR

Ken Valentine is Fluor's Global Excellence Leader for Control Systems and a registered chemical engineer at the Sugar Land, Tex., office.

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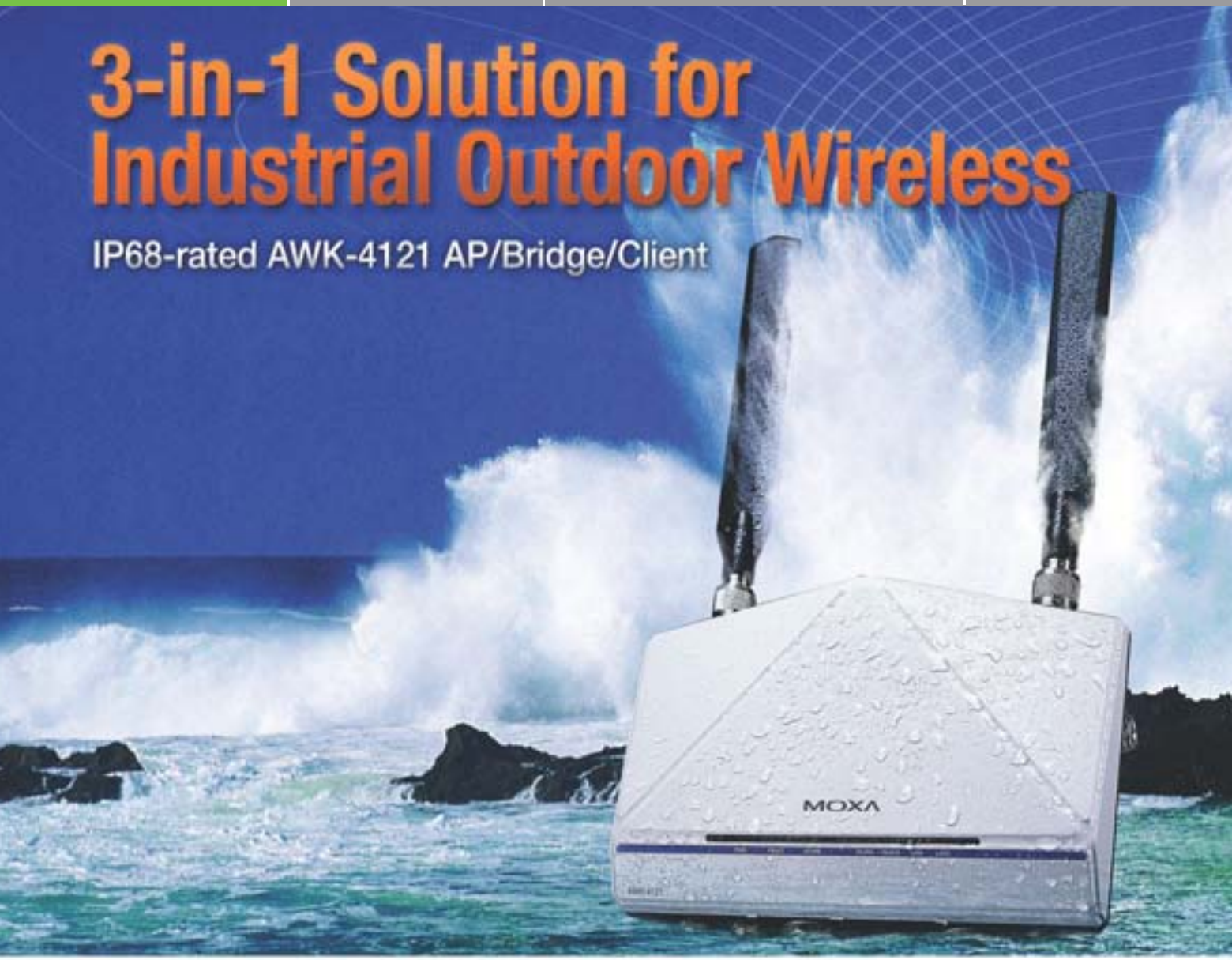
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